MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

RECEIVER, INTERFACE, DIGITAL, KEY CODE JHK

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic interface receiver module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.477 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be JHK.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/73-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

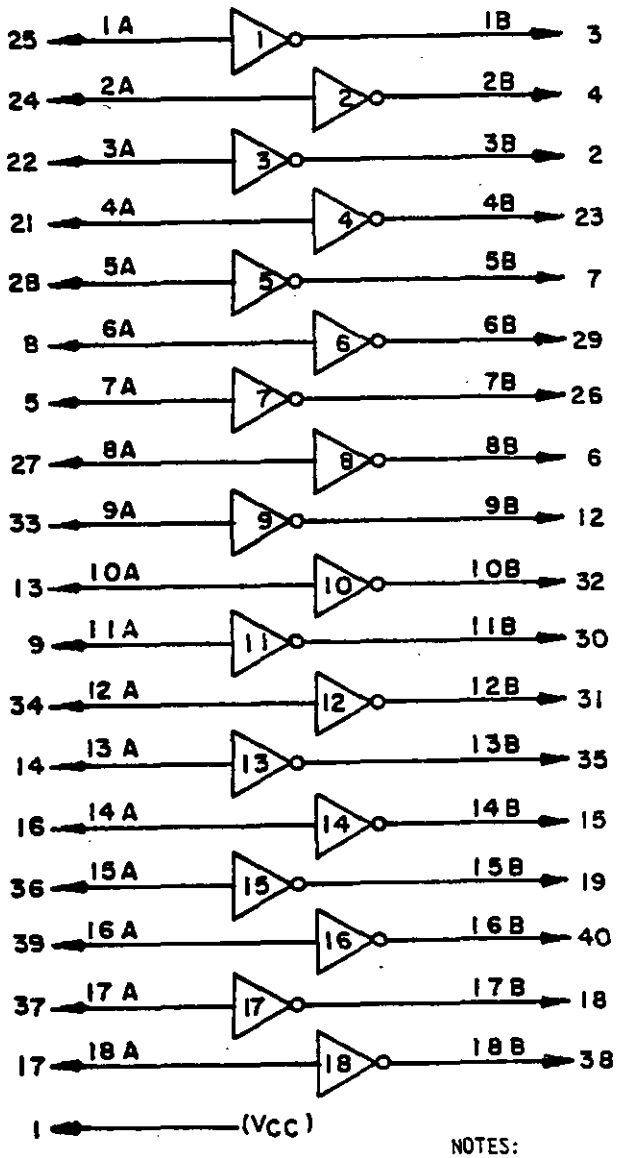
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. 1A-18A = receiver inputs.
2. 1B-18B = receiver outputs.
3. V_{CC} = 5 V dc ±10%.
4. GND = 0 volt reference.
5. Each input is tied internally to V_{CC} through a 6.81 kΩ ±5% pullup resistor.

FIGURE 1. JHK module function diagram, interface receiver.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A3	Isolation	IV	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B18	V_{OUT} (low-loaded)	V	S	0.01	1.00	0.01	1.00	0.01	1.00	0.01	1.00	V
C1-C18	Input breakdown current	VI	S	0	2	0	2	0	2	0	2	mA
D1-D18	Output short circuit current	VII	X	-20	-55	-20	-55	-20	-55	-20	-55	mA
E1-E18	Output leakage current	VIII	X	0	1	0	1	0	1	0	1	mA
F1-F18	Input clamp diode voltage	IX	X	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	V
G1-G18	Output voltage, low level	X	X	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
H1-H18	Output voltage, high level	XI	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J18	Input current, high level	XII	X	330	492	330	492	330	492	330	492	µA
K1-K18	Input current, low level	XIII	X	0.72	2.00	0.72	2.00	0.72	2.00	0.72	2.00	mA
L1-L18	Input threshold margin check (V_{CC} = 4.5 V)	XIV	S	0.4	1.0	0.4	1.0	0.4	1.0	0.4	1.0	V
L19-L36	Input threshold margin check (V_{CC} = 5.5 V)	XIV	S	0.4	1.2	0.4	1.2	0.4	1.2	0.4	1.2	V
M1	Power supply current, low-level outputs	XV	S	120	370	120	370	120	370	120	370	mA
M2	Power supply current, high-level outputs	XV	S	110	350	110	350	110	350	110	350	mA
N1-N18	Transition time (t_{TLH}), C_L = 50 pF	XVI	S	10	29	10	29	10	42	10	42	ns
N19-N36	Transition time (t_{TLH}), C_L = 150 pF	XVI	Q	24	58	24	58	22	80	22	80	ns
P1-P18	Transition time (t_{THL}), C_L = 50 pF	XVII	S	3	14	3	14	3	17	3	17	ns

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P19-P36	Transition time (t_{THL}), C_L = 150 pF	XVII	Q	3	17	3	17	3	20	3	20	ns
Q1-Q18	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	5	35	5	35	5	37	5	37	ns
Q19-Q36	Propagation delay time (t_{PLH}), C_L = 150 pF	XVIII	Q	5	40	5	40	5	42	5	42	ns
R1-R18	Propagation delay time (t_{PHL}), C_L = 50 pF	XIX	S	5	35	5	35	5	37	5	37	ns
R19-R36	Propagation delay time (t_{PHL}), C_L = 150 pF	XIX	Q	5	40	5	40	5	42	5	42	ns
S1-S18	Output crosstalk, low level	XX	Q	0	800	0	800	0	800	0	800	mV
S19-S36	Output crosstalk, low level	XX	Q	0	500	0	500	0	500	0	500	mV
T1-T18	Output crosstalk, high level	XXI	Q	0	400	0	400	0	400	0	400	mV
U	Power supply transient amplitude	XXII	Q	0	400	0	400	0	400	0	400	mV
V	Power supply filter capacitance	XXIII	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
W1-W18	Input capacitance	XXIV	Q	2	40	2	40	2	40	2	40	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing: Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through W shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition V, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through V shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through W shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/73-1 supersedes part number M28787/73. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits of table I.

Maximum power dissipation	
All outputs at low level	2,035 mW (outputs not loaded)
All outputs at high level	2,035 mW (outputs not loaded)
Supply voltage	5.0 V dc ±10 percent
Input voltage	-0.5 V minimum, 5.5 V maximum
Input level threshold	
Negative going (V_t-)	1.2 V minimum, 1.6 V maximum
Positive going (V_t+)	1.8 V minimum, 2.2 V maximum

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Input threshold margin	0.4 V minimum
Input current, low level	
V_{IL} = 0.0 V	2.00 mA maximum
V_{IL} = 0.5 V	1.8 mA maximum
Input current, high level	
V_{IH} = 2.4 V	492 µA maximum
V_{IH} = 5.5 V	2.0 mA maximum
Output voltage	
Low level (I_{OL} = 20 mA)	0.5 V maximum
High level (I_{OH} = -800 µA)	2.4 V minimum
Output drive current maximum capability	
Low level (V_{OL} = 0.5 V)	20 mA
High level (V_{OH} = 2.4 V)	-800 µA
Propagation delay time (t_{PHL})	
50 pF load	5 ns minimum, 37 ns maximum
150 pF load	5 ns minimum, 42 ns maximum
Propagation delay time (t_{PLH})	
50 pF load	5 ns minimum, 37 ns maximum
150 pF load	5 ns minimum, 42 ns maximum

6.5 Module function. This module contains 18 independent interface receivers. It is intended for use as a receiver when transmitting logic level signals from chassis to chassis within an equipment cabinet. When flat cable is used for interconnection, the length shall be less than 7 feet, and the driver and receiver shall be located adjacent to the ends of the cable. A dedicated driver shall be used to drive the interface receiver. A dedicated driver is defined as a circuit that drives no other circuits, either on its own module, or on another module except the interface receiver module.

6.5.1 Input to output relationship. Each interface receiver has the input to output relationship of table II for positive logic.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

6.5.3 Input hysteresis. Each input has the hysteresis characteristics of table III.

6.5.4 Input threshold margin. Each input has a positive-going threshold voltage (V_t+) that is at least 0.4 volt greater than its negative-going threshold voltage (V_t-).

6.5.5 Crosstalk relationship. The module has the following crosstalk relationships.

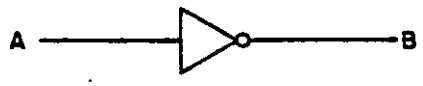
6.5.5.1 Module crosstalk. When connected in the worst case mode of operation (all outputs but one switching simultaneously from high level to low level), the JHK module could exhibit substantial crosstalk. When the quiet output is a low level, there may be as much as an 800 mV positive-going spike coincident with the negative edge of the switching outputs.

6.5.5.2 Circuit crosstalk. The crosstalk on any output pin is nearly proportional to the number of switching outputs. For every switching output in group A (circuits 1-4), the noise induced on a low-level quiet output in group A is as much as 103 mV. For every switching output outside group A (circuits 5-18), the noise induced on a low-level quiet output in group A is as much as 35 mV. The preceding relationships also hold for group B (circuits 5-8), group C (circuits 9-12), group D (circuits 13-14), and group E (circuits 15-18).

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TABLE II. Input to output relationship. 1/, 2/

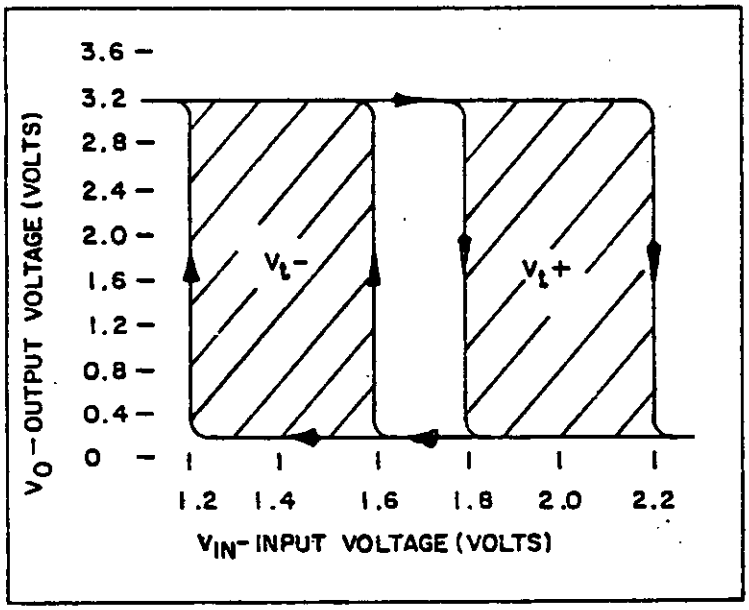
INPUT	OUTPUT
A	B
L	H
H	L



$B = \overline{A}$

1/ H = High-voltage level.
2/ L = Low-voltage level.

TABLE III. Input hysteresis characteristics.



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6.6 0 volt reference. The reference will be zero volt for the test setup and not necessarily earth ground.

6.7 Decoupling. Decoupling of test voltages and V_{CC} to minimize noise in the test setup is permissible except test condition U, power supply transient amplitude test.

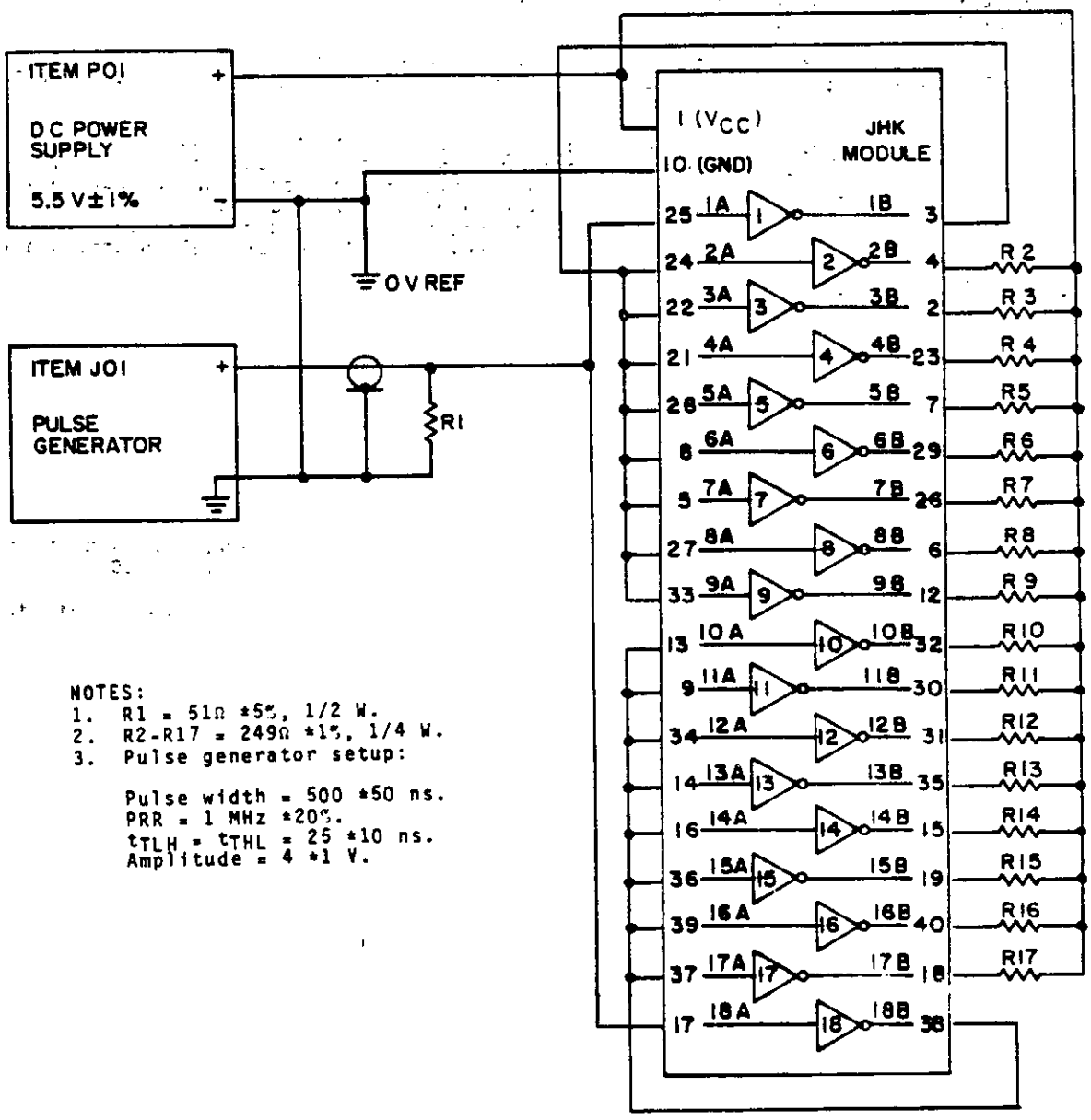
6.8 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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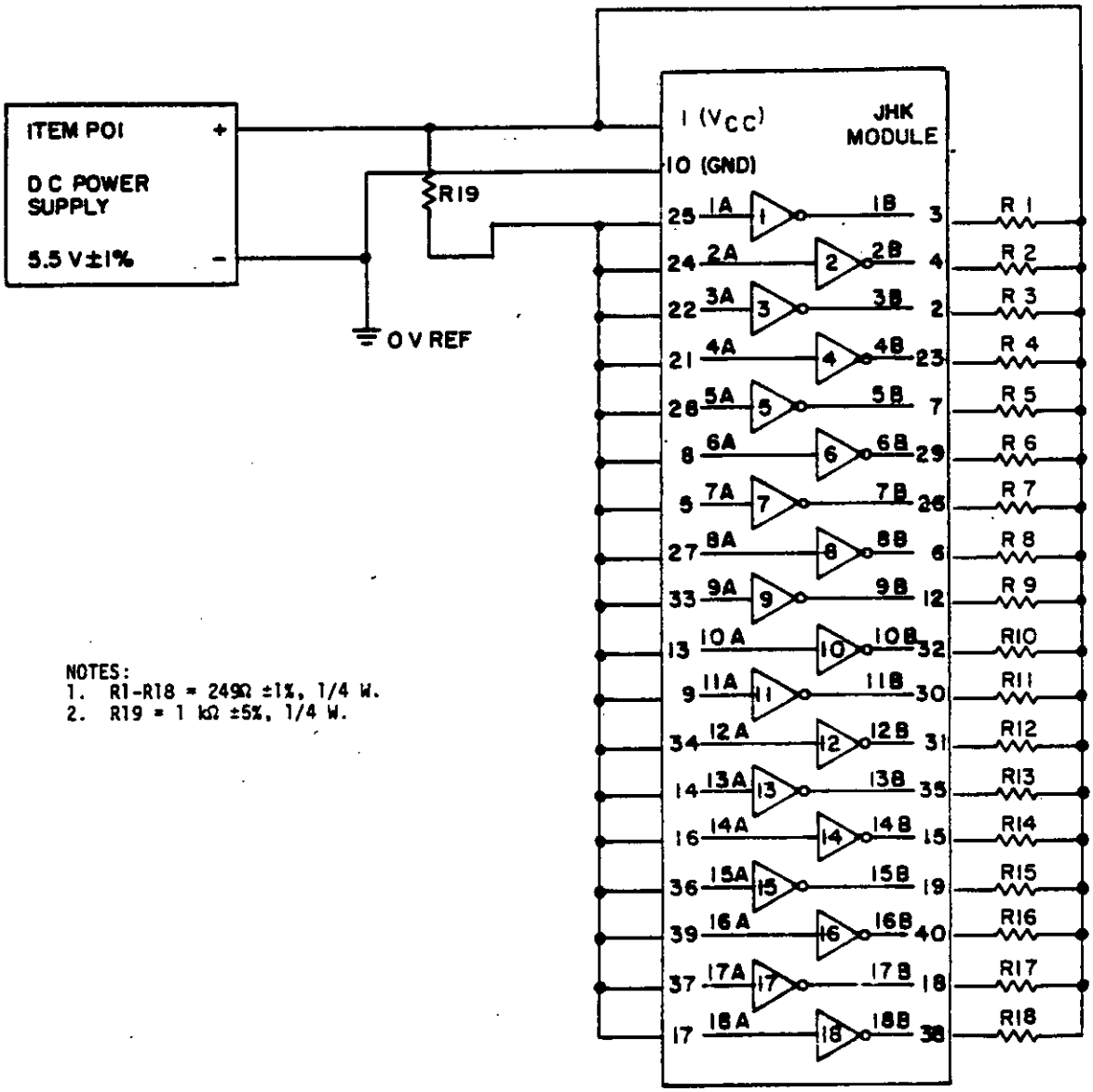


NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. R2-R17 = 249Ω ±1%, 1/4 W.
3. Pulse generator setup:

   Pulse width = 500 ±50 ns.
   PRR = 1 MHz ±20%.
   $t_{TLH} = t_{THL}$ = 25 ±10 ns.
   Amplitude = 4 ±1 V.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1-R18 = 249Ω ±1%, 1/4 W.
2. R19 = 1 kΩ ±5%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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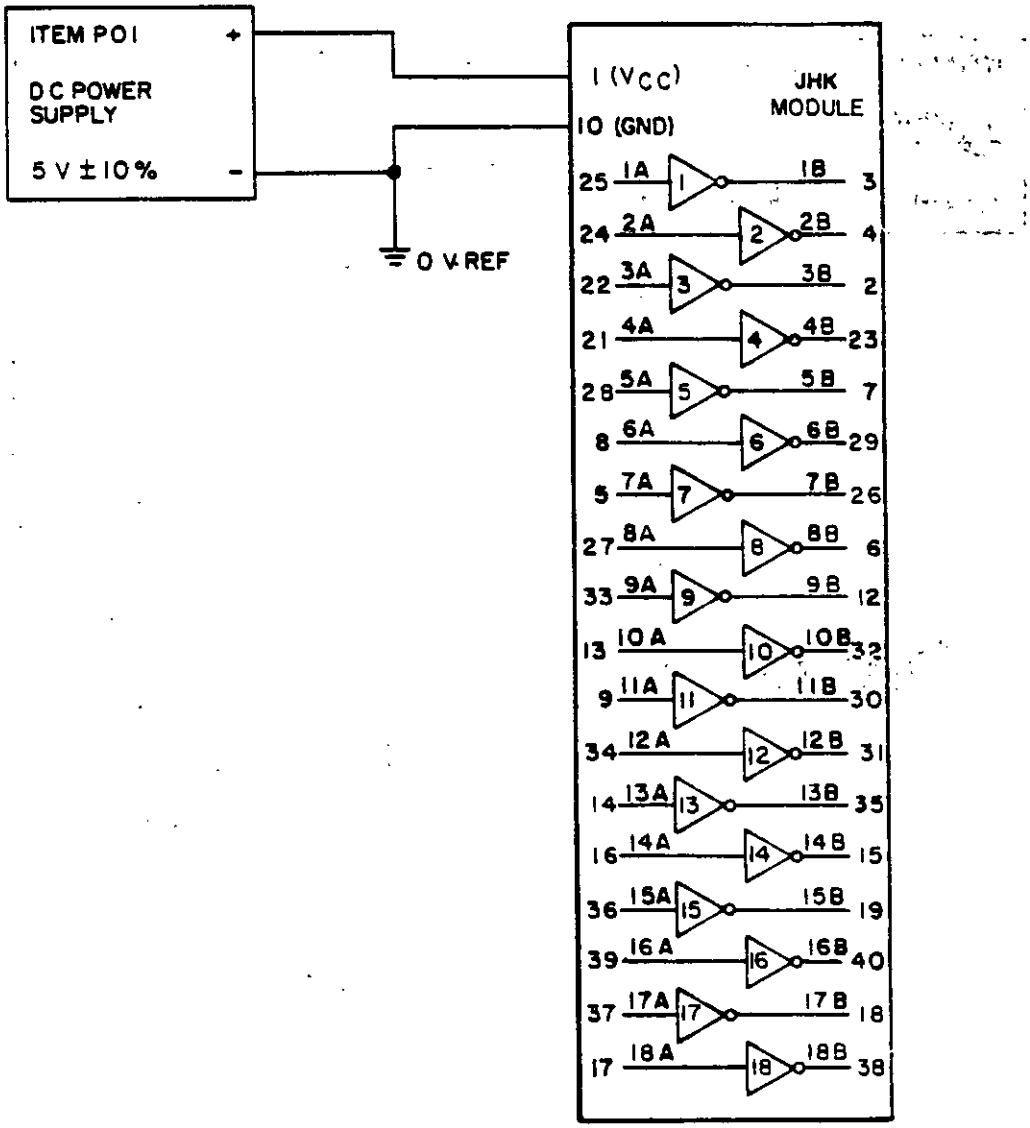


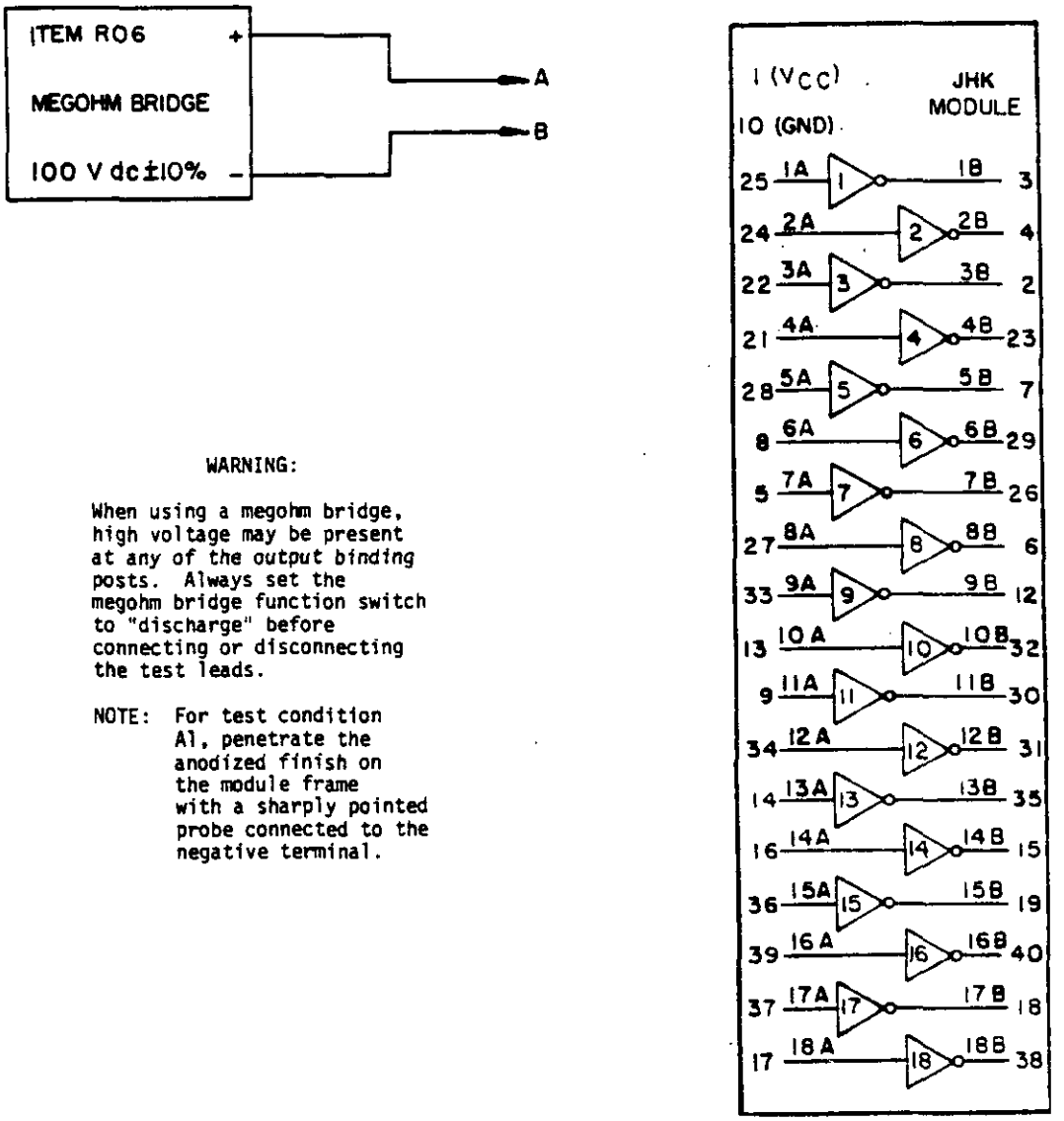
FIGURE 4. Vibration test circuit.

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TABLE IV. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	All pins	Frame	10	∞	MΩ	5
A2	"	All pins except 11 and 20	11,20	"	"	"	"
A3	"	11	20	"	"	"	"

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition A1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

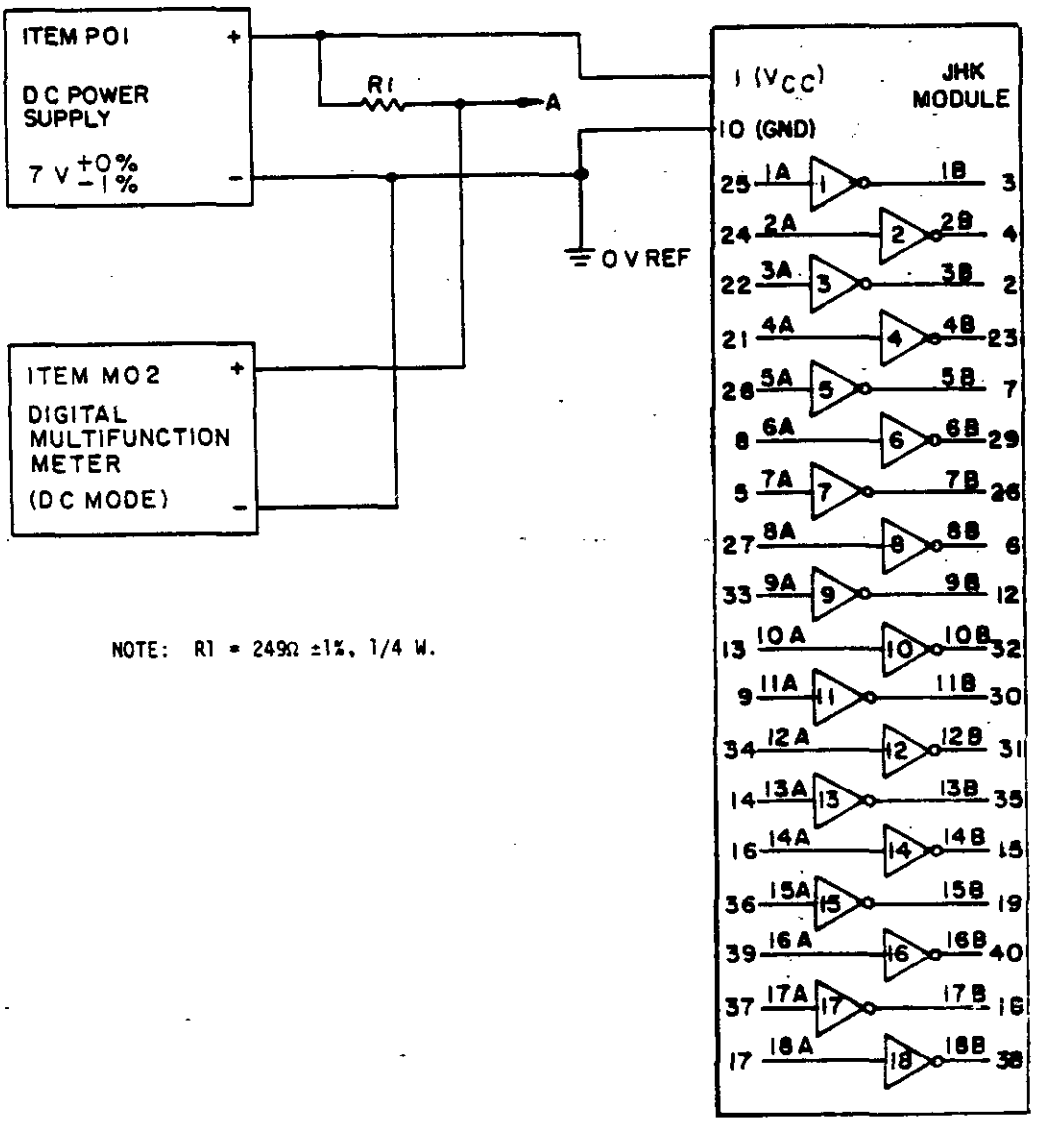
FIGURE 5. Isolation test circuit for table IV.

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TABLE V. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
B1	1B V_{OUT} (low-loaded)	3	0.01	1.00	V	6
B2	2B "	4	"	"	"	"
B3	3B "	2	"	"	"	"
B4	4B "	23	"	"	"	"
B5	5B "	7	"	"	"	"
B6	6B "	29	"	"	"	"
B7	7B "	26	"	"	"	"
B8	8B "	6	"	"	"	"
B9	9B "	12	"	"	"	"
B10	10B "	32	"	"	"	"
B11	11B "	30	"	"	"	"
B12	12B "	31	"	"	"	"
B13	13B "	35	"	"	"	"
B14	14B "	15	"	"	"	"
B15	15B "	19	"	"	"	"
B16	16B "	40	"	"	"	"
B17	17B "	18	"	"	"	"
B18	18B "	38	"	"	"	"

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NOTE: R1 = 249Ω ±1%, 1/4 W.

* FIGURE 6. V_{OUT} (low-loaded) test circuit for table V.

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TABLE VI. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1A Input breakdown current	25	5,8,9,13,14,16,17,21,22, 24,27,28,33,34,36,37,39	0	2	mA	7
C2	2A "	24	5,8,9,13,14,16,17,21,22, 25,27,28,33,34,36,37,39	"	"	"	"
C3	3A "	22	5,8,9,13,14,16,17,21,24, 25,27,28,33,34,36,37,39	"	"	"	"
C4	4A "	21	5,8,9,13,14,16,17,22,24, 25,27,28,33,34,36,37,39	"	"	"	"
C5	5A "	28	5,8,9,13,14,16,17,21,22, 24,25,27,33,34,36,37,39	"	"	"	"
C6	6A "	8	5,9,13,14,16,17,21,22, 24,25,27,28,33,34,36,37, 39	"	"	"	"
C7	7A "	5	8,9,13,14,16,17,21,22, 24,25,27,28,33,34,36,37, 39	"	"	"	"
C8	8A "	27	5,8,9,13,14,16,17,21,22, 24,25,28,33,34,36,37, 39	"	"	"	"
C9	9A "	33	5,8,9,13,14,16,17,21,22, 24,25,27,28,34,36,37,39	"	"	"	"
C10	10A "	13	5,8,9,14,16,17,21,22,24, 25,27,28,33,34,36,37,39	"	"	"	"
C11	11A "	9	5,8,13,14,16,17,21,22, 24,25,27,28,33,34,36,37, 39	"	"	"	"
C12	12A "	34	5,8,9,13,14,16,17,21,22, 24,25,27,28,33,36,37,39	"	"	"	"
C13	13A "	14	5,8,9,13,16,17,21,22,24, 25,27,28,33,34,36,37,39	"	"	"	"

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TABLE VI. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C14	14A Input breakdown current	16	5,8,9,13,14,17,21,22,24, 25,27,28,33,34,36,37,39	0	2	mA	7
C15	15A "	36	5,8,9,13,14,16,17,21,22, 24,25,27,28,33,34,37,39	"	"	"	"
C16	16A "	39	5,8,9,13,14,16,17,21,22, 24,25,27,28,33,34,36,37	"	"	"	"
C17	17A "	37	5,8,9,13,14,16,17,21,22, 24,25,27,28,33,34,36,39	"	"	"	"
C18	18A "	17	5,8,9,13,14,16,21,22,24, 25,27,28,33,34,36,37,39	"	"	"	"

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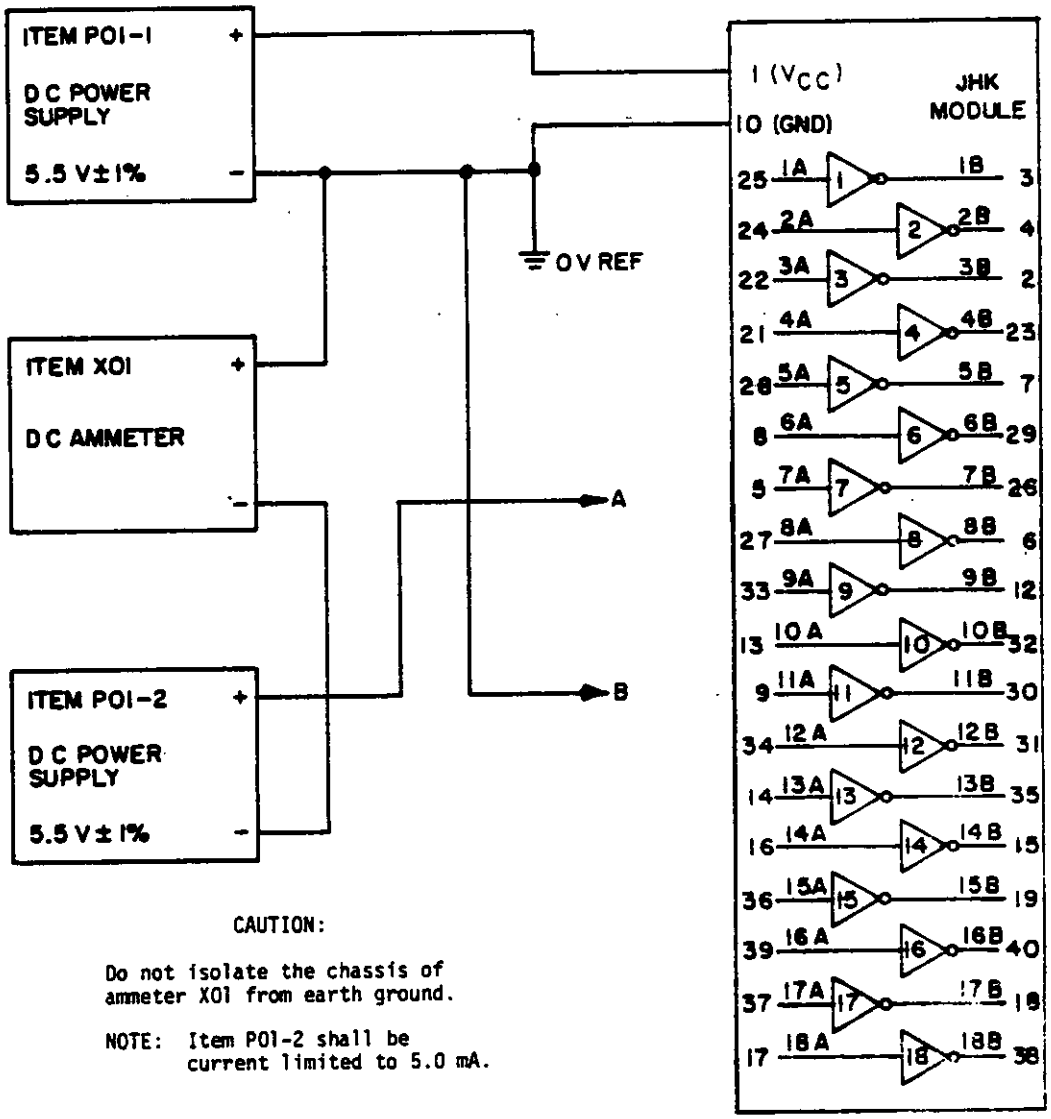


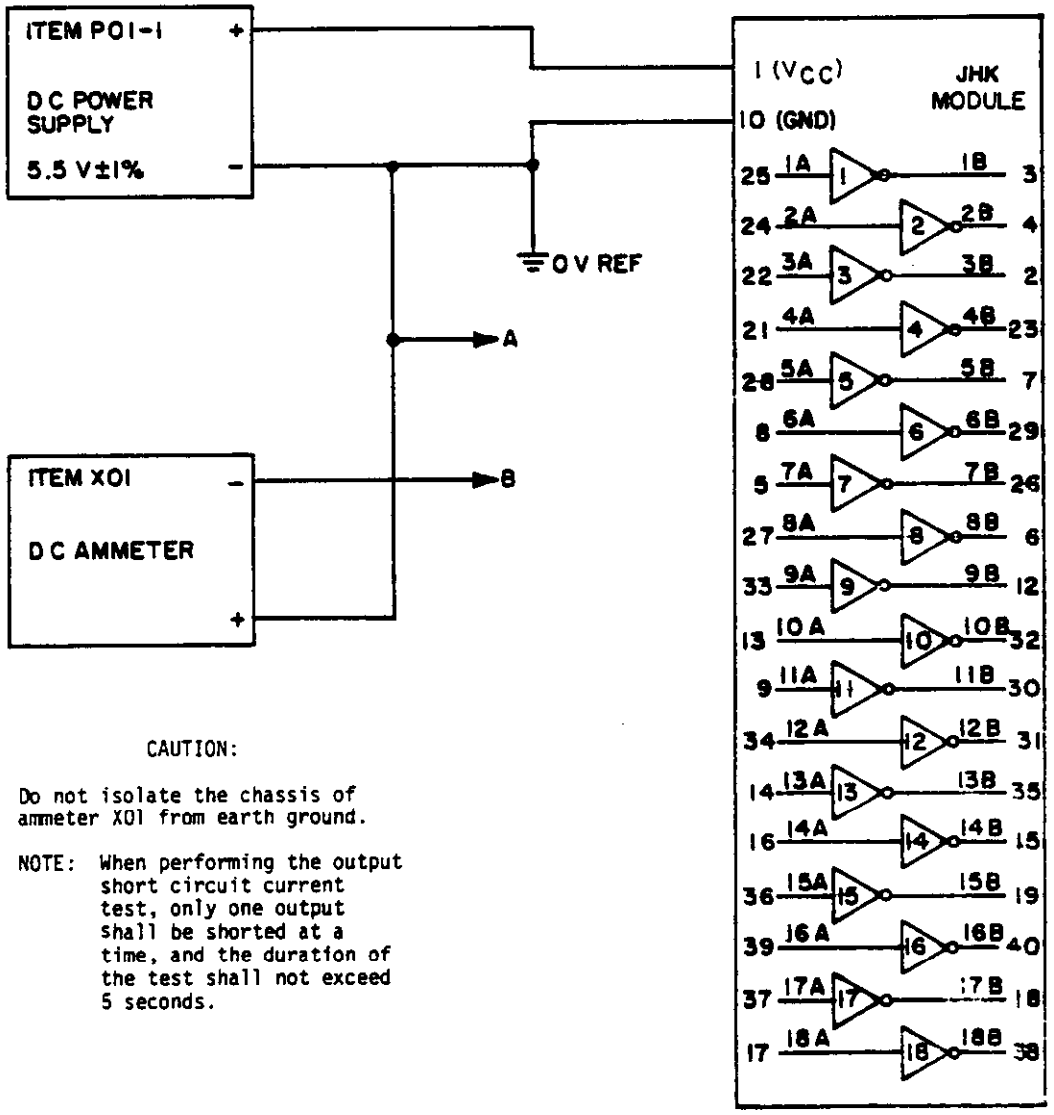
FIGURE 7. Input breakdown current test circuit for table VI.

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TABLE VII. Output short circuit current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	1B Output short circuit current	5,8,9,14,16,17, 21,22,24,25,27, 28,33,34,36,37, 39,13	3	-20	-55	mA	8
D2	2B "	"	4	"	"	"	"
D3	3B "	"	2	"	"	"	"
D4	4B "	"	23	"	"	"	"
D5	5B "	"	7	"	"	"	"
D6	6B "	"	29	"	"	"	"
D7	7B "	"	26	"	"	"	"
D8	8B "	"	6	"	"	"	"
D9	9B "	"	12	"	"	"	"
D10	10B "	"	32	"	"	"	"
D11	11B "	"	30	"	"	"	"
D12	12B "	"	31	"	"	"	"
D13	13B "	"	35	"	"	"	"
D14	14B "	"	15	"	"	"	"
D15	15B "	"	19	"	"	"	"
D16	16B "	"	40	"	"	"	"
D17	17B "	"	18	"	"	"	"
D18	18B "	"	38	"	"	"	"

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTE: When performing the output short circuit current test, only one output shall be shorted at a time, and the duration of the test shall not exceed 5 seconds.

FIGURE 8. Output short circuit current test circuit for table VII.

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TABLE VIII. Output leakage current test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
E1	1B Output leakage current	3	0	1	mA	9
E2	2B "	4	"	"	"	"
E3	3B "	2	"	"	"	"
E4	4B "	23	"	"	"	"
E5	5B "	7	"	"	"	"
E6	6B "	29	"	"	"	"
E7	7B "	26	"	"	"	"
E8	8B "	6	"	"	"	"
E9	9B "	12	"	"	"	"
E10	10B "	32	"	"	"	"
E11	11B "	30	"	"	"	"
E12	12B "	31	"	"	"	"
E13	13B "	35	"	"	"	"
E14	14B "	15	"	"	"	"
E15	15B "	19	"	"	"	"
E16	16B "	40	"	"	"	"
E17	17B "	18	"	"	"	"
E18	18B "	38	"	"	"	"

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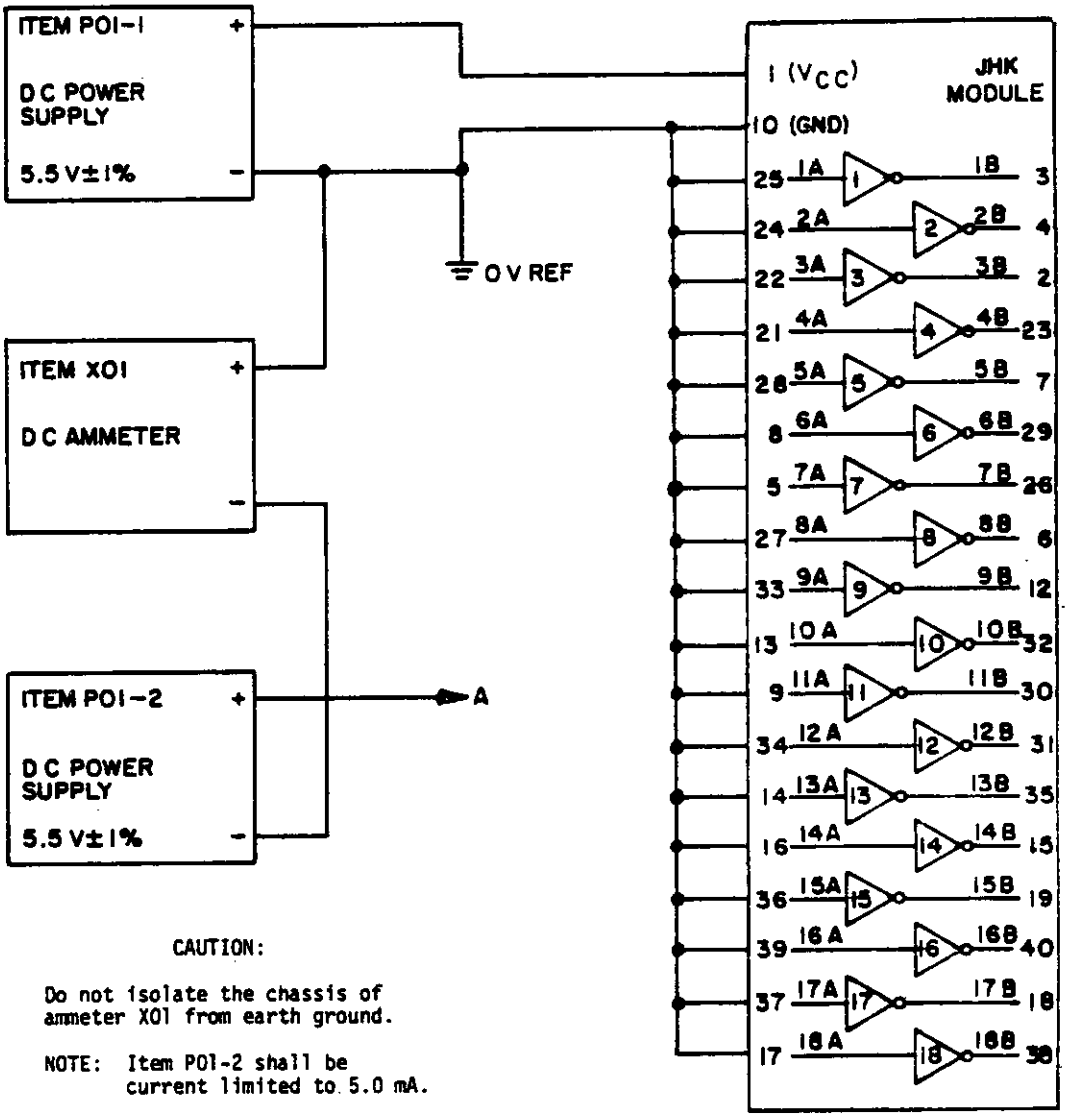


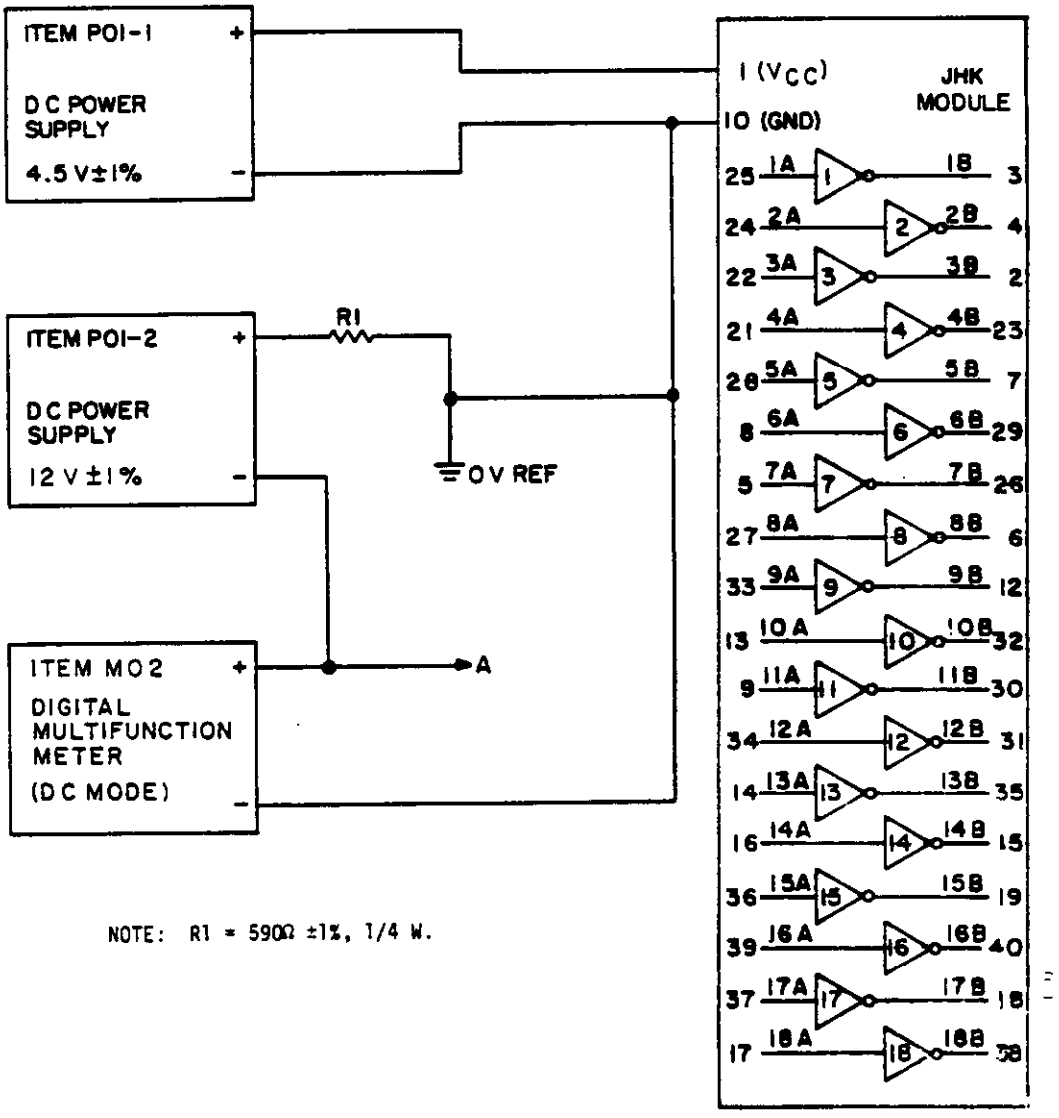
FIGURE 9. Output leakage current test circuit for table VIII.

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TABLE IX. Input clamp diode voltage test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
F1	1A Input clamp diode voltage	25	-0.4	-1.2	V	10
F2	2A "	24	"	"	"	"
F3	3A "	22	"	"	"	"
F4	4A "	21	"	"	"	"
F5	5A "	28	"	"	"	"
F6	6A "	8	"	"	"	"
F7	7A "	5	"	"	"	"
F8	8A "	27	"	"	"	"
F9	9A "	33	"	"	"	"
F10	10A "	13	"	"	"	"
F11	11A "	9	"	"	"	"
F12	12A "	34	"	"	"	"
F13	13A "	14	"	"	"	"
F14	14A "	16	"	"	"	"
F15	15A "	36	"	"	"	"
F16	16A "	39	"	"	"	"
F17	17A "	37	"	"	"	"
F18	18A "	17	"	"	"	"

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NOTE: R1 = 590Ω ±1%, 1/4 W.

* FIGURE 10. Input clamp diode voltage test circuit for table IX.

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TABLE X. Output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1B Output voltage, low level	25	Open	3	0.0	0.5	V	11
G2	2B "	24	"	4	"	"	"	"
G3	3B "	22	"	2	"	"	"	"
G4	4B "	21	"	23	"	"	"	"
G5	5B "	28	"	7	"	"	"	"
G6	6B "	8	"	29	"	"	"	"
G7	7B "	5	"	26	"	"	"	"
G8	8B "	27	"	6	"	"	"	"
G9	9B "	33	"	12	"	"	"	"
G10	10B "	13	"	32	"	"	"	"
G11	11B "	9	"	30	"	"	"	"
G12	12B "	34	"	31	"	"	"	"
G13	13B "	14	"	35	"	"	"	"
G14	14B "	16	"	15	"	"	"	"
G15	15B "	36	"	19	"	"	"	"
G16	16B "	39	"	40	"	"	"	"
G17	17B "	37	"	18	"	"	"	"
G18	18B "	17	"	38	"	"	"	"

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TABLE XI. Output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1B Output voltage, high level	24,22,21, 28,8,5,27, 33,13,9,34, 14,16,36, 39,37,17	25	3	2.4	4.5	V	11
H2	2B "	25,22,21, 28,8,5,27, 33,13,9,34, 14,16,36, 39,37,17	24	4	"	"	"	"
H3	3B "	25,24,21, 28,8,5,27, 33,13,9,34, 14,16,36, 39,37,17	22	2	"	"	"	"
H4	4B "	25,24,22, 28,8,5,27, 33,13,9,34, 14,16,36, 39,37,17	21	23	"	"	"	"
H5	5B "	25,24,22, 21,8,5,27, 33,13,9,34, 14,16,36, 39,37,17	28	7	"	"	"	"
H6	6B "	25,24,22, 21,28,5,27, 33,13,9,34, 14,16,36, 39,37,17	8	29	"	"	"	"
H7	7B "	25,24,22, 21,28,8,27, 33,13,9,34, 14,16,36, 39,37,17	5	26	"	"	"	"

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TABLE XI. Output voltage (high-level) test - Continued.

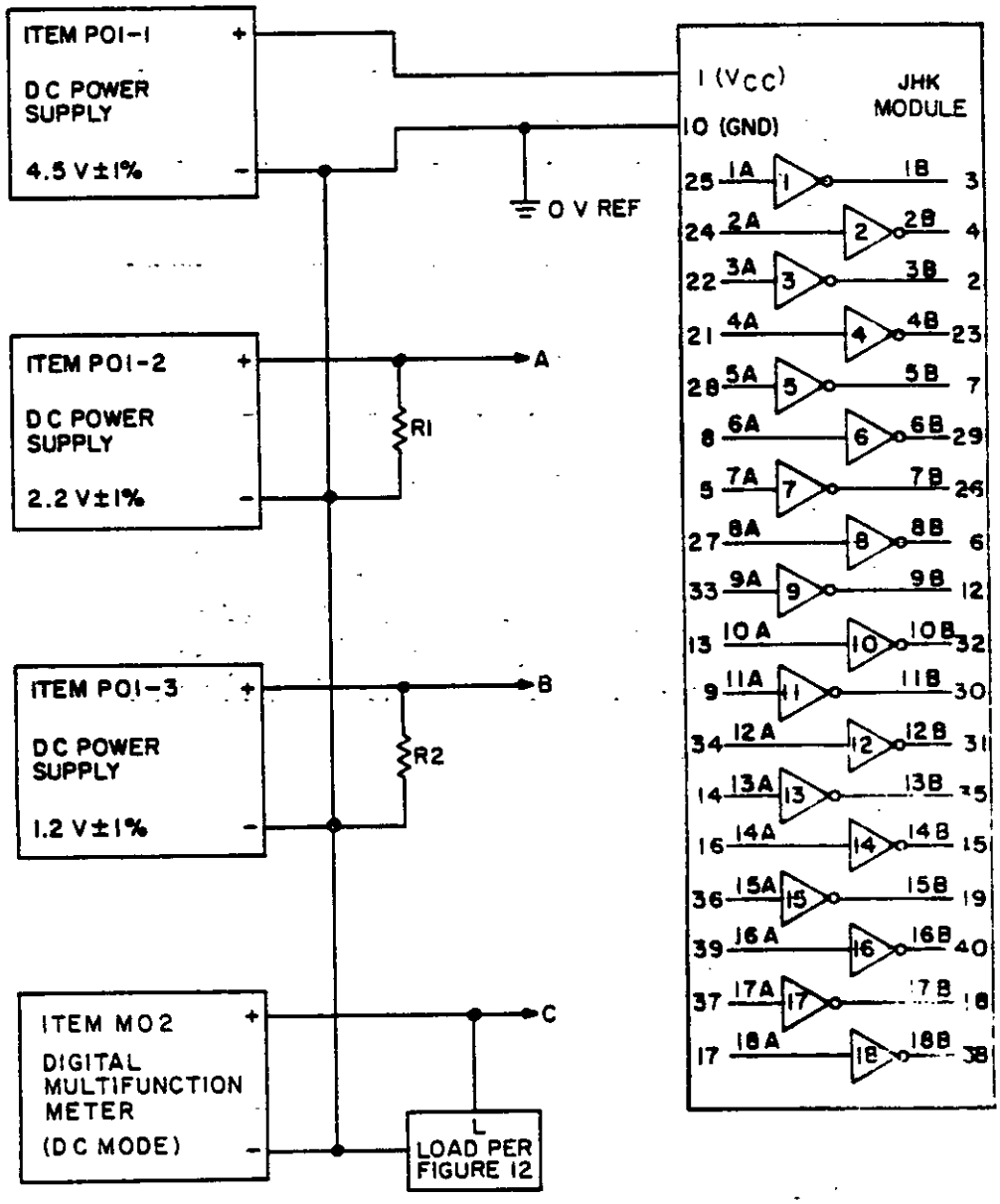
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H8	8B Output voltage, high level	25,24,22, 21,28,8,5, 33,13,9,34, 14,16,36, 39,37,17	27	6	2.4	4.5	V	11
H9	9B "	25,24,22, 21,28,8,5, 27,13,9,34, 14,16,36, 39,37,17	33	12	"	"	"	"
H10	10B "	25,24,22, 21,28,8,5, 27,33,9,34, 14,16,36, 39,37,17	13	32	"	"	"	"
H11	11B "	25,24,22, 21,28,8,5, 27,33,13, 34,14,16, 36,39,37,17	9	30	"	"	"	"
H12	12B "	25,24,22, 21,28,8,5, 27,33,13, 9,14,16, 36,39,37,17	34	31	"	"	"	"
H13	13B "	25,24,22, 21,28,8,5, 27,33,13, 9,34,16, 36,39,37,17	14	35	"	"	"	"
H14	14B "	25,24,22, 21,28,8,5, 27,33,13, 9,34,14, 36,39,37,17	16	15	"	"	"	"

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TABLE XI. Output voltage (high-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H15	15B	Output voltage, high level	25,24,22, 21,28,8,5, 27,33,13,9, 34,14,16, 39,37,17	36	19	2.4	4.5	V	11
H16	16B	"	25,24,22, 21,28,8,5, 27,33,13,9, 34,14,16, 36,37,17	39	40	"	"	"	"
H17	17B	"	25,24,22, 21,28,8,5, 27,33,13,9, 34,14,16, 36,39,17	37	18	"	"	"	"
H18	18B	"	25,24,22, 21,28,8,5, 27,33,13,9, 34,14,16, 36,39,37	17	38	"	"	"	"

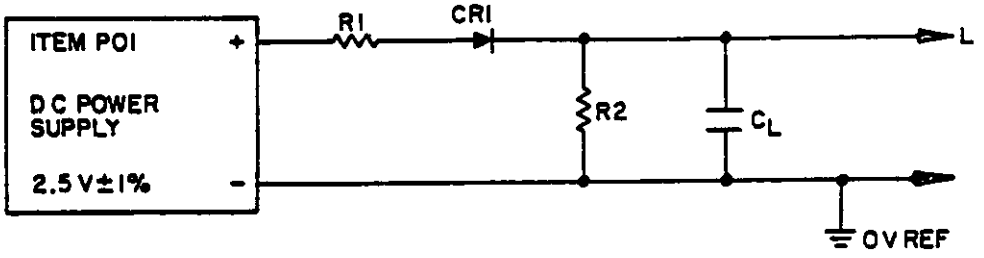
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NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

* FIGURE 11. Output voltage (low-level and high-level) test circuit for tables X and XI.

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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 20 mA ±5% with point L held at 0.5 V.
3. R2 = 3.01 kΩ ±1%, 1/8 W.
4. Test type S:

   C_L = 50 ±5 pF (including probe and parasitic capacitance).

   Test type Q:

   C_L = 150 ±5 pF (including probe and parasitic capacitance) except for crosstalk test which will use 50 ±5 pF.

5. Capacitor C_L is not required for dc electrical test conditions G1-G18 and H1-H18.
6. Lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 mm).

* FIGURE 12. JHK logic load circuit.

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TABLE XII. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	1A Input current, high level	25	5,8,9,13,14,16,17,21,22, 24,27,28,33,34,36,37,39	330	492	µA	13
J2	2A "	24	5,8,9,13,14,16,17,21,22, 25,27,28,33,34,36,37,39	"	"	"	"
J3	3A "	22	5,8,9,13,14,16,17,21,24, 25,27,28,33,34,36,37,39	"	"	"	"
J4	4A "	21	5,8,9,13,14,16,17,22,24, 25,27,28,33,34,36,37,39	"	"	"	"
J5	5A "	28	5,8,9,13,14,16,17,21,22, 24,25,27,33,34,36,37,39	"	"	"	"
J6	6A "	8	5,9,13,14,16,17,21,22,24, 25,27,28,33,34,36,37,39	"	"	"	"
J7	7A "	5	8,9,13,14,16,17,21,22,24, 25,27,28,33,34,36,37,39	"	"	"	"
J8	8A "	27	5,8,9,13,14,16,17,21,22, 24,25,28,33,34,36,37,39	"	"	"	"
J9	9A "	33	5,8,9,13,14,16,17,21,22, 24,25,27,28,34,36,37,39	"	"	"	"
J10	10A "	13	5,8,9,14,16,17,21,22,24, 25,27,28,33,34,36,37,39	"	"	"	"
J11	11A "	9	5,8,13,14,16,17,21,22,24, 25,27,28,33,34,36,37,39	"	"	"	"
J12	12A "	34	5,8,9,13,14,16,17,21,22, 24,25,27,28,33,36,37,39	"	"	"	"
J13	13A "	14	5,8,9,13,16,17,21,22,24, 25,27,28,33,34,36,37,39	"	"	"	"
J14	14A "	16	5,8,9,13,14,17,21,22,24, 25,27,28,33,34,36,37,39	"	"	"	"

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TABLE XII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J15	15A Input current, high level	36	5,8,9,13,14,16,17,21,22, 24,25,27,28,33,34,37,39	330	492	µA	13
J16	16A "	39	5,8,9,13,14,16,17,21,22, 24,25,27,28,33,34,36,37	"	"	"	"
J17	17A "	37	5,8,9,13,14,16,17,21,22, 24,25,27,28,33,34,36,39	"	"	"	"
J18	18A "	17	5,8,9,13,14,16,21,22,24, 25,27,28,33,34,36,37,39	"	"	"	"

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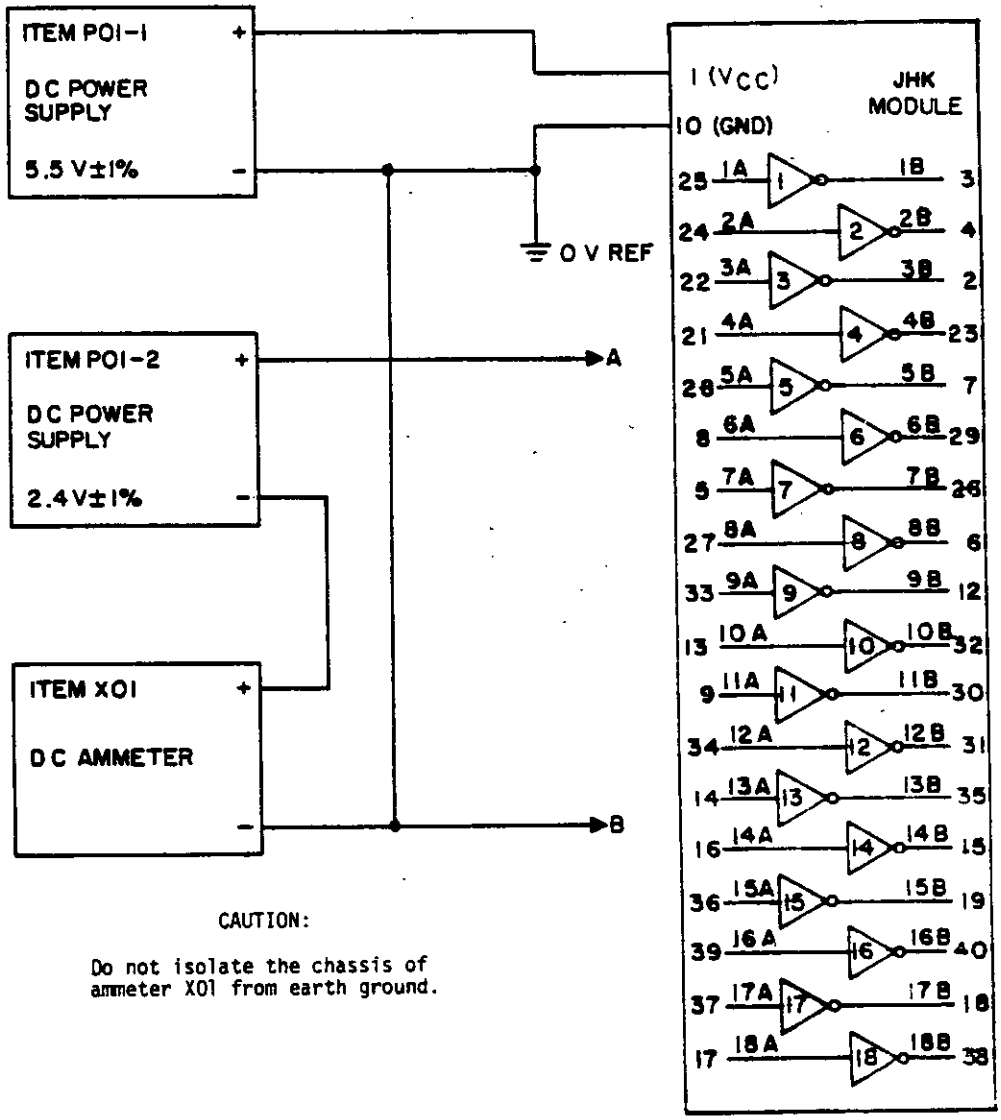


FIGURE 13. Input current (high-level) test circuit for table XII.

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*TABLE XIII. Input current (low-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
K1	1A Input current, low level	25	0.72	2.00	mA	14
K2	2A "	24	"	"	"	"
K3	3A "	22	"	"	"	"
K4	4A "	21	"	"	"	"
K5	5A "	28	"	"	"	"
K6	6A "	8	"	"	"	"
K7	7A "	5	"	"	"	"
K8	8A "	27	"	"	"	"
K9	9A "	33	"	"	"	"
K10	10A "	13	"	"	"	"
K11	11A "	9	"	"	"	"
K12	12A "	34	"	"	"	"
K13	13A "	14	"	"	"	"
K14	14A "	16	"	"	"	"
K15	15A "	36	"	"	"	"
K16	16A "	39	"	"	"	"
K17	17A "	37	"	"	"	"
K18	18A "	17	"	"	"	"

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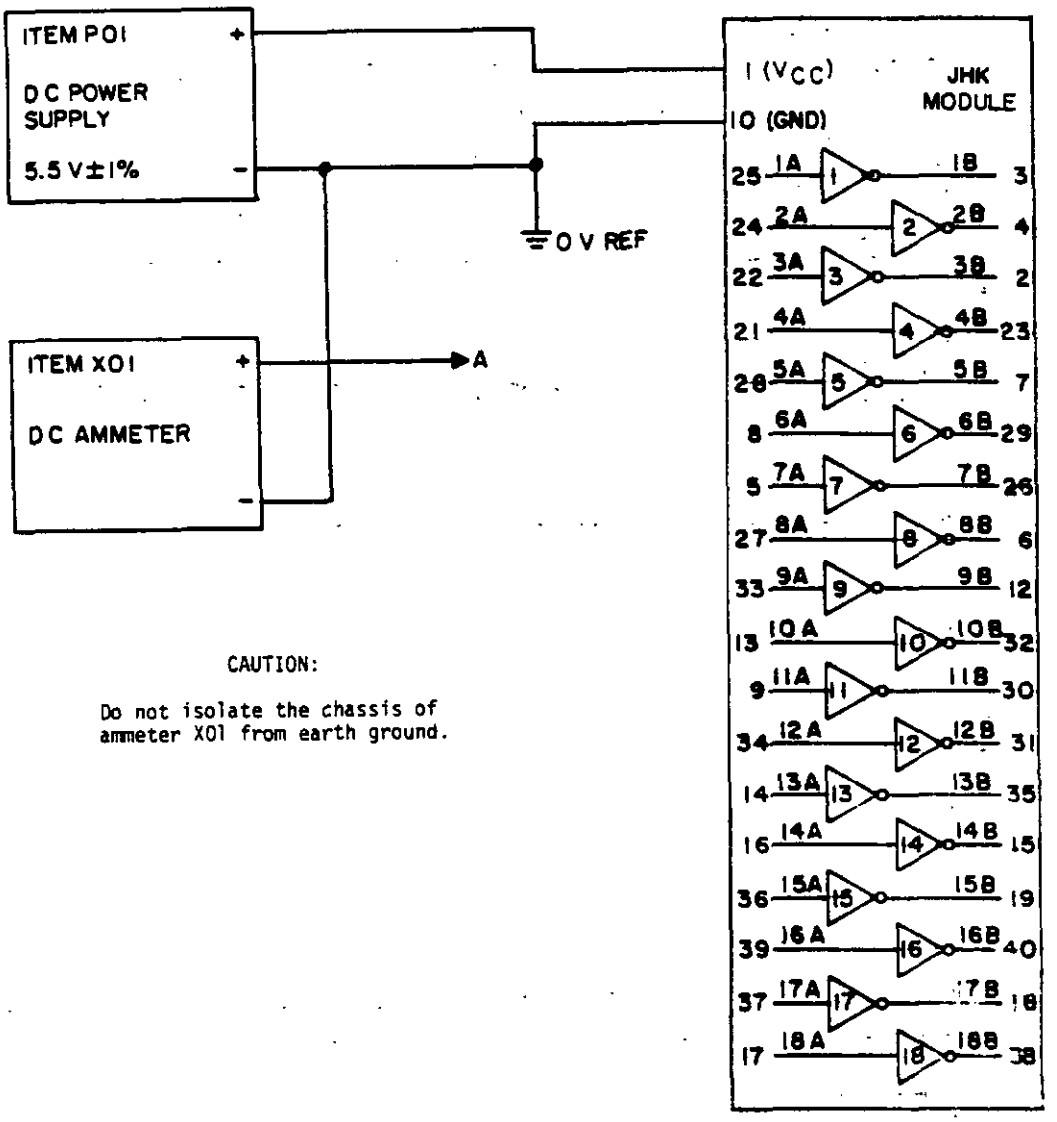


FIGURE 14. Input current (low-level) test circuit for table XIII.

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*TABLE XIV. Input threshold margin check test.

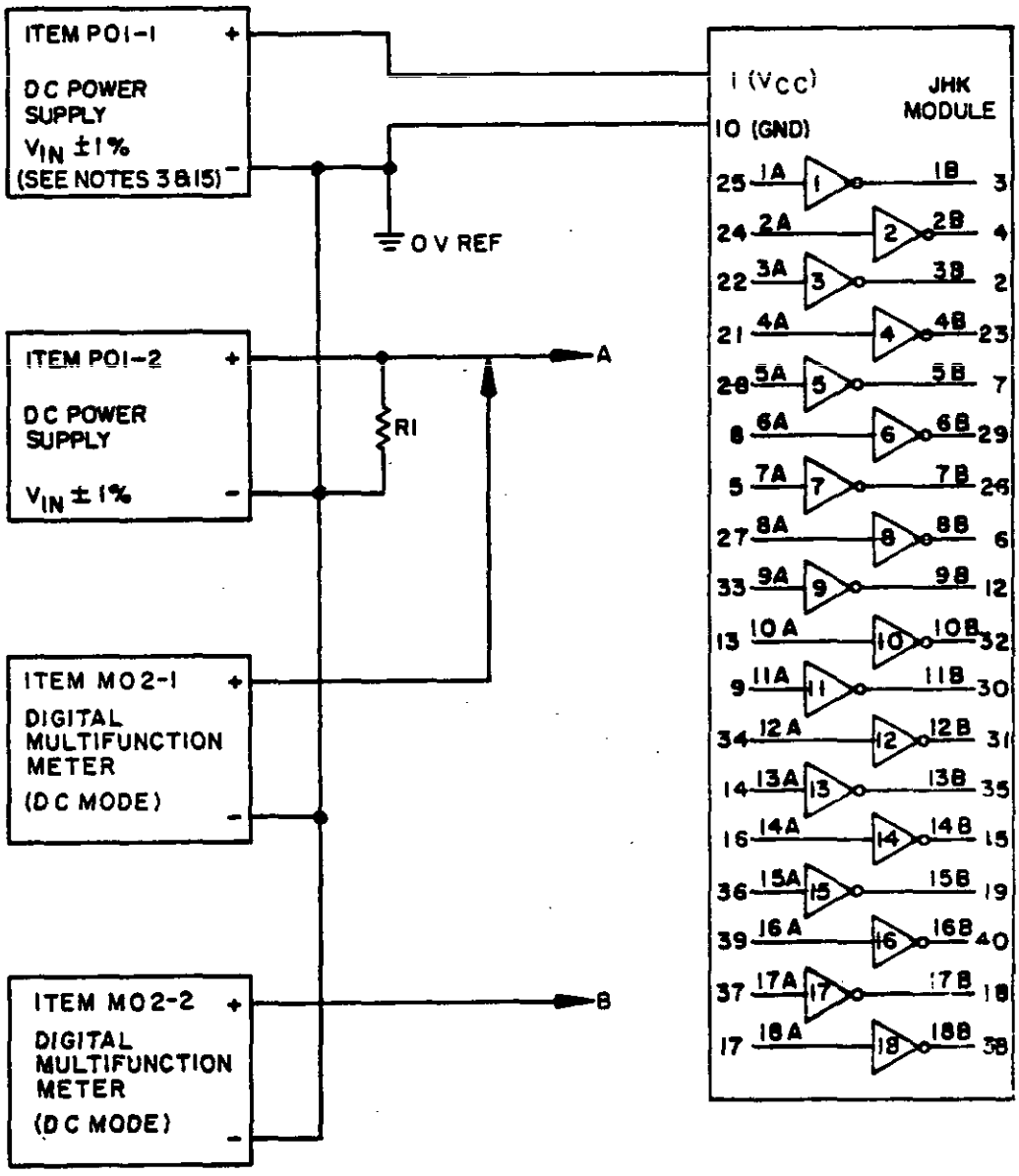
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	1A Input threshold margin check, V_{CC} = 4.5 V	25	3	0.4	1.0	V	15
L2	2A "	24	4	"	"	"	"
L3	3A "	22	2	"	"	"	"
L4	4A "	21	23	"	"	"	"
L5	5A "	28	7	"	"	"	"
L6	6A "	8	29	"	"	"	"
L7	7A "	5	26	"	"	"	"
L8	8A "	27	6	"	"	"	"
L9	9A "	33	12	"	"	"	"
L10	10A "	13	32	"	"	"	"
L11	11A "	9	30	"	"	"	"
L12	12A "	34	31	"	"	"	"
L13	13A "	14	35	"	"	"	"
L14	14A "	16	15	"	"	"	"
L15	15A "	36	19	"	"	"	"
L16	16A "	39	40	"	"	"	"
L17	17A "	37	18	"	"	"	"
L18	18A "	17	38	"	"	"	"
L19	1A Input threshold margin check, V_{CC} = 5.5 V	25	3	"	1.2	"	"
L20	2A Input threshold margin check, V_{CC} = 5.5 V	24	4	"	1.2	"	"

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*TABLE XIV. Input threshold margin check test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L21	3A Input threshold margin check, V_{CC} = 5.5 V	22	2	0.4	1.2	V	15
L22	4A "	21	23	"	"	"	"
L23	5A "	28	7	"	"	"	"
L24	6A "	8	29	"	"	"	"
L25	7A "	5	26	"	"	"	"
L26	8A "	27	6	"	"	"	"
L27	9A "	33	12	"	"	"	"
L28	10A "	13	32	"	"	"	"
L29	11A "	9	30	"	"	"	"
L30	12A "	34	31	"	"	"	"
L31	13A "	14	35	"	"	"	"
L32	14A "	16	15	"	"	"	"
L33	15A "	36	19	"	"	"	"
L34	16A "	39	40	"	"	"	"
L35	17A "	37	18	"	"	"	"
L36	18A "	17	38	"	"	"	"

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* FIGURE 15. Input threshold margin check test circuit for table XIV.

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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Output loading per figure 12 is optional.
3. Adjust item P01-1 to 4.5 V.
4. Adjust item P01-2 to 1.2 V.
5. Connect points A and B to module pins.
6. Observe item M02-2; verify that output is a high level (2.4 V minimum, 4.5 V maximum).
7. While observing item M02-2, adjust item P01-2 to raise V_{IN} until the output switches to a low level (0.0 V minimum, 0.5 V maximum).
8. Measure V_{IN} on item M02-1 and record measured voltage as V_A.
9. Adjust item P01-2 until V_{IN} is 2.2 V.
10. Verify that the output being monitored is a low level.
11. While observing item M02-2 adjust item P01-2 to lower V_{IN} until the output switches to a high level.
12. Measure V_{IN} on item M02-2 and record measured voltage as V_B.
13. The result of V_A-V_B shall comply with the requirements of table XIV.
14. Repeat notes 4 through 14 for test conditions L2 through L18.
15. Adjust item P01-1 to 5.5 V.
16. Repeat notes 4 through 14 for test conditions L19 through L36.

* FIGURE 15. Input threshold margin check test circuit for table XIV - Continued.

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TABLE XV. Power supply current (low-level outputs and high-level outputs) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Power supply current, low-level outputs	C	Open	120	370	mA	16
M2	Power supply current, high-level outputs	Open	C	110	350	mA	16

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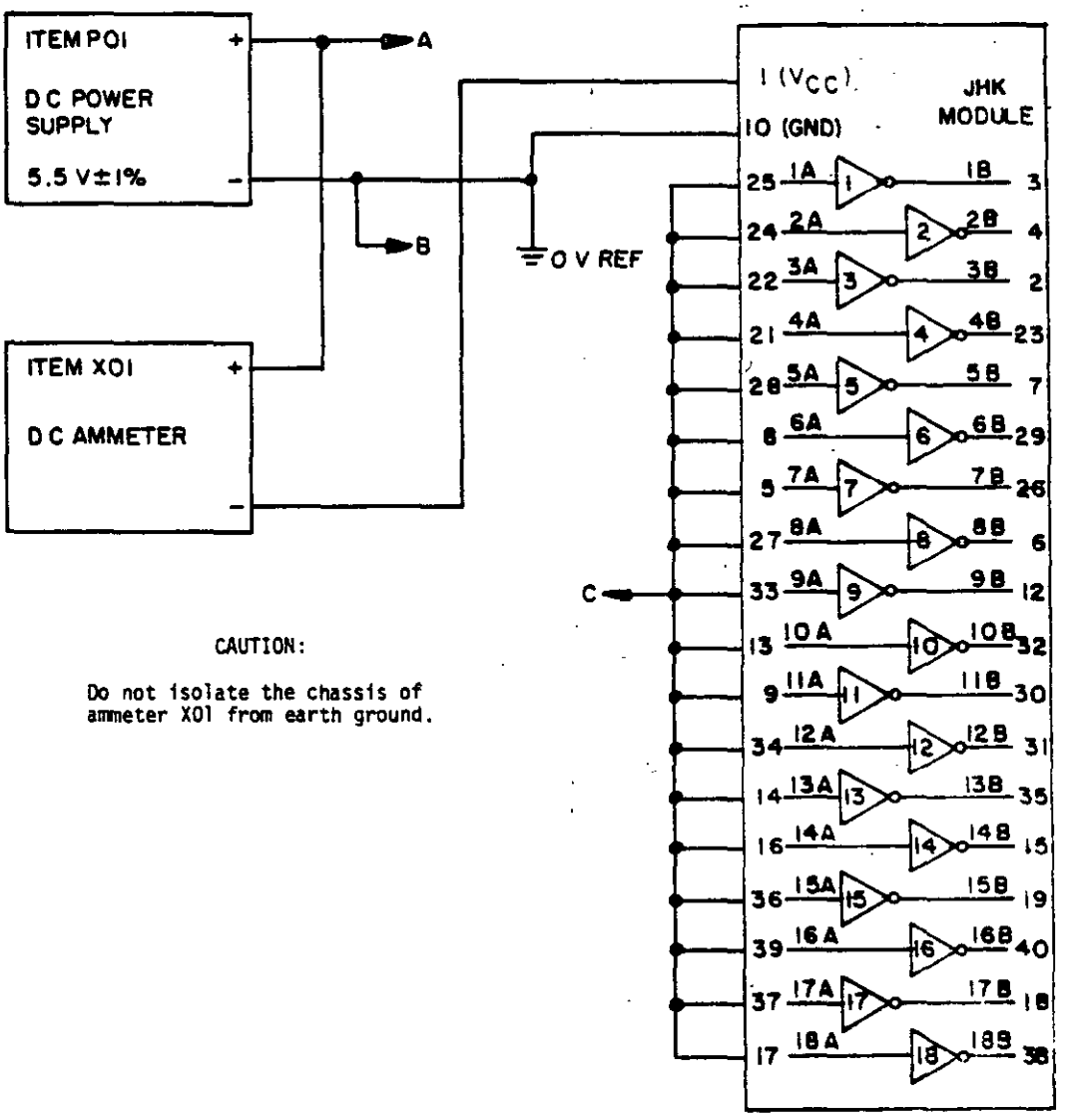


FIGURE 16. Power supply current (low-level outputs and high-level outputs) test circuit for table XV.

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TABLE XVI. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N1	1B Transition time (t_{TLH}), C_L = 50 pF	25	3	10	29	ns	17,18
N2	2B "	24	4	"	"	"	"
N3	3B "	22	2	"	"	"	"
N4	4B "	21	23	"	"	"	"
N5	5B "	28	7	"	"	"	"
N6	6B "	8	29	"	"	"	"
N7	7B "	5	26	"	"	"	"
N8	8B "	27	6	"	"	"	"
N9	9B "	33	12	"	"	"	"
N10	10B "	13	32	"	"	"	"
N11	11B "	9	30	"	"	"	"
N12	12B "	34	31	"	"	"	"
N13	13B "	14	35	"	"	"	"
N14	14B "	16	15	"	"	"	"
N15	15B "	36	19	"	"	"	"
N16	16B "	39	40	"	"	"	"
N17	17B "	37	18	"	"	"	"
N18	18B "	17	38	"	"	"	"
N19	1B Transition time (t_{TLH}), C_L = 150 pF	25	3	24	58	"	"
N20	2B Transition time (t_{TLH}), C_L = 150 pF	24	4	24	58	"	"

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TABLE XVI. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N21	3B Transition time (t_{TLH}), C_L = 150 pF	22	2	24	58	ns	17,18
N22	4B "	21	23	"	"	"	"
N23	5B "	28	7	"	"	"	"
N24	6B "	8	29	"	"	"	"
N25	7B "	5	26	"	"	"	"
N26	8B "	27	6	"	"	"	"
N27	9B "	33	12	"	"	"	"
N28	10B "	13	32	"	"	"	"
N29	11B "	9	30	"	"	"	"
N30	12B "	34	31	"	"	"	"
N31	13B "	14	35	"	"	"	"
N32	14B "	16	15	"	"	"	"
N33	15B "	36	19	"	"	"	"
N34	16B "	39	40	"	"	"	"
N35	17B "	37	18	"	"	"	"
N36	18B "	17	38	"	"	"	"

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TABLE XVII. Transition time (t_{THL}) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P1	1B Transition time (t_{THL}), C_L = 50 pF	25	3	3	14	ns	17,18
P2	2B "	24	4	"	"	"	"
P3	3B "	22	2	"	"	"	"
P4	4B "	21	23	"	"	"	"
P5	5B "	28	7	"	"	"	"
P6	6B "	8	29	"	"	"	"
P7	7B "	5	26	"	"	"	"
P8	8B "	27	6	"	"	"	"
P9	9B "	33	12	"	"	"	"
P10	10B "	13	32	"	"	"	"
P11	11B "	9	30	"	"	"	"
P12	12B "	34	31	"	"	"	"
P13	13B "	14	35	"	"	"	"
P14	14B "	16	15	"	"	"	"
P15	15B "	36	19	"	"	"	"
P16	16B "	39	40	"	"	"	"
P17	17B "	37	18	"	"	"	"
P18	18B "	17	38	"	"	"	"
P19	1B Transition time (t_{THL}), C_L = 150 pF	25	3	"	17	"	"
P20	2B Transition time (t_{THL}), C_L = 150 pF	24	4	"	17	"	"

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TABLE XVII. Transition time (t_{THL}) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P21	3B Transition time (t_{THL}), C_L = 150 pF	22	2	3	17	ns	17,18
P22	4B "	21	23	"	"	"	"
P23	5B "	28	7	"	"	"	"
P24	6B "	8	29	"	"	"	"
P25	7B "	5	26	"	"	"	"
P26	8B "	27	6	"	"	"	"
P27	9B "	33	12	"	"	"	"
P28	10B "	13	32	"	"	"	"
P29	11B "	9	30	"	"	"	"
P30	12B "	34	31	"	"	"	"
P31	13B "	14	35	"	"	"	"
P32	14B "	16	15	"	"	"	"
P33	15B "	36	19	"	"	"	"
P34	16B "	39	40	"	"	"	"
P35	17B "	37	18	"	"	"	"
P36	18B "	17	38	"	"	"	"

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TABLE XVIII. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Q1	1B Propagation delay time (t_{PLH}), C_L = 50 pF	25	3	5	35	ns	17,18
Q2	2B "	24	4	"	"	"	"
Q3	3B "	22	2	"	"	"	"
Q4	4B "	21	23	"	"	"	"
Q5	5B "	28	7	"	"	"	"
Q6	6B "	8	29	"	"	"	"
Q7	7B "	5	26	"	"	"	"
Q8	8B "	27	6	"	"	"	"
Q9	9B "	33	12	"	"	"	"
Q10	10B "	13	32	"	"	"	"
Q11	11B "	9	30	"	"	"	"
Q12	12B "	34	31	"	"	"	"
Q13	13B "	14	35	"	"	"	"
Q14	14B "	16	15	"	"	"	"
Q15	15B "	36	19	"	"	"	"
Q16	16B "	39	40	"	"	"	"
Q17	17B "	37	18	"	"	"	"
Q18	18B "	17	38	"	"	"	"
Q19	1B Propagation delay time (t_{PLH}), C_L = 150 pF	25	3	"	40	"	"
Q20	2B Propagation delay time (t_{PLH}), C_L = 150 pF	24	4	"	40	"	"

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TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Q21	3B Propagation delay time (t_{PLH}), C_L = 150 pF	22	2	5	40	ns	17,18
Q22	4B "	21	23	"	"	"	"
Q23	5B "	28	7	"	"	"	"
Q24	6B "	8	29	"	"	"	"
Q25	7B "	5	26	"	"	"	"
Q26	8B "	27	6	"	"	"	"
Q27	9B "	33	12	"	"	"	"
Q28	10B "	13	32	"	"	"	"
Q29	11B "	9	30	"	"	"	"
Q30	12B "	34	31	"	"	"	"
Q31	13B "	14	35	"	"	"	"
Q32	14B "	16	15	"	"	"	"
Q33	15B "	36	19	"	"	"	"
Q34	16B "	39	40	"	"	"	"
Q35	17B "	37	18	"	"	"	"
Q36	18B "	17	38	"	"	"	"

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TABLE XIX. Propagation delay time (t_{PHL}) test.

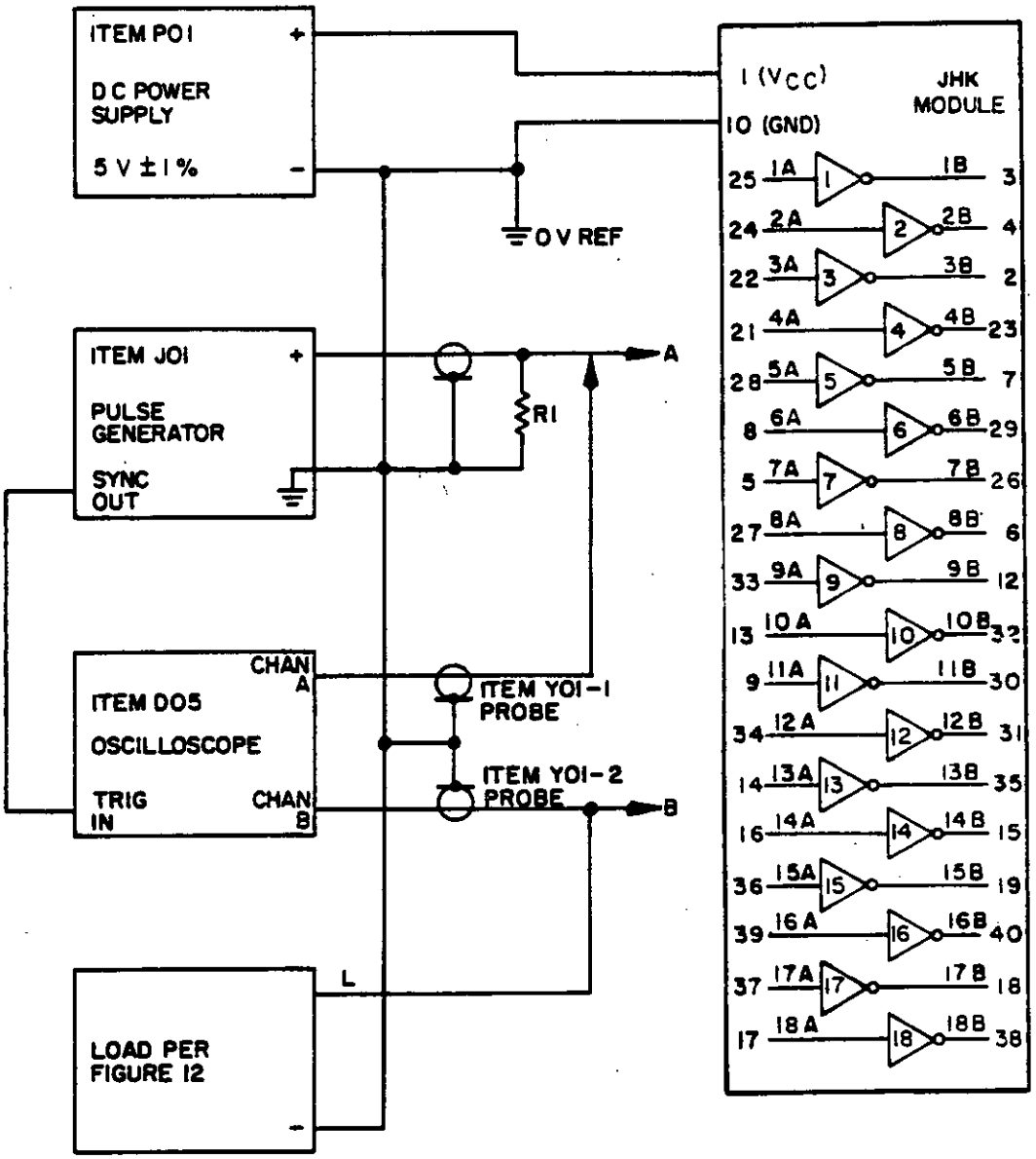
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
R1	1B Propagation delay time (t_{PHL}), C_L = 50 pF	25	3	5	35	ns	17,18
R2	2B "	24	4	"	"	"	"
R3	3B "	22	2	"	"	"	"
R4	4B "	21	23	"	"	"	"
R5	5B "	28	7	"	"	"	"
R6	6B "	8	29	"	"	"	"
R7	7B "	5	26	"	"	"	"
R8	8B "	27	6	"	"	"	"
R9	9B "	33	12	"	"	"	"
R10	10B "	13	32	"	"	"	"
R11	11B "	9	30	"	"	"	"
R12	12B "	34	31	"	"	"	"
R13	13B "	14	35	"	"	"	"
R14	14B "	16	15	"	"	"	"
R15	15B "	36	19	"	"	"	"
R16	16B "	39	40	"	"	"	"
R17	17B "	37	18	"	"	"	"
R18	18B "	17	38	"	"	"	"
R19	1B Propagation delay time (t_{PHL}), C_L = 150 pF	25	3	"	40	"	"
R20	2B Propagation delay time (t_{PHL}), C_L = 150 pF	24	4	"	40	"	"

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TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
R21	3B Propagation delay time (t_{PHL}), C_L = 150 pF	22	2	5	40	ns	17,18
R22	4B "	21	23	"	"	"	"
R23	5B "	28	7	"	"	"	"
R24	6B "	8	29	"	"	"	"
R25	7B "	5	26	"	"	"	"
R26	8B "	27	6	"	"	"	"
R27	9B "	33	12	"	"	"	"
R28	10B "	13	32	"	"	"	"
R29	11B "	9	30	"	"	"	"
R30	12B "	34	31	"	"	"	"
R31	13B "	14	35	"	"	"	"
R32	14B "	16	15	"	"	"	"
R33	15B "	36	19	"	"	"	"
R34	16B "	39	40	"	"	"	"
R35	17B "	37	18	"	"	"	"
R36	18B "	17	38	"	"	"	"

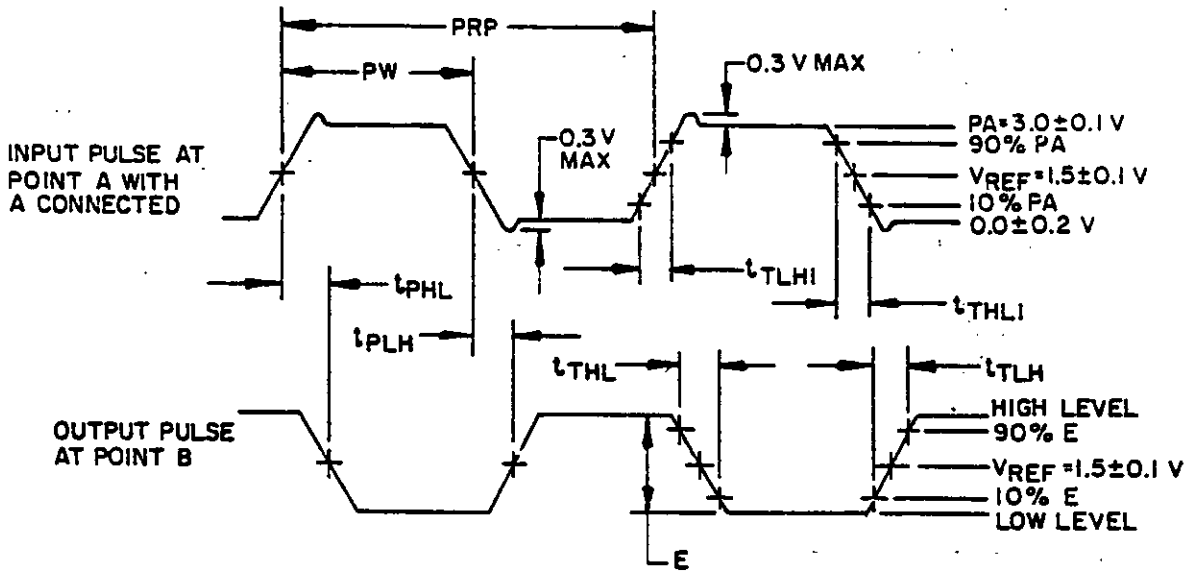
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NOTE: R1 = 51.1Ω ±1%, 1/4 W.

FIGURE 17. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XVI, XVII, XVIII, and XIX.

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NOTES:
1. Input pulse parameters:

   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.

   PW = 100 ±10 ns.
   PRP = 500 ±50 ns.

2. Before making the measurements specified in table XVI verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between resistor R1 and input pin; between probes Y01-1 and Y01-2 and input and output pins; and between load circuit and output pin, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

* FIGURE 18. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XVI, XVII, XVIII, and XIX.

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TABLE XX. Output crosstalk (low-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	1B Output crosstalk, low level	25	5,8,9,13, 14,16,17, 21,22,24, 27,28,33, 34,36,37, 39	3	Open	0	800	mV	19,20
S2	2B "	24	5,8,9,13, 14,16,17, 21,22,25, 27,28,33, 34,36,37, 39	4	"	"	"	"	"
S3	3B "	22	5,8,9,13, 14,16,17, 21,24,25, 27,28,33, 34,36,37, 39	2	"	"	"	"	"
S4	4B "	21	5,8,9,13, 14,16,17, 22,24,25, 27,28,33, 34,36,37, 39	23	"	"	"	"	"
S5	5B "	28	5,8,9,13, 14,16,17, 21,22,24, 25,27,33, 34,36,37, 39	7	"	"	"	"	"
S6	6B "	8	5,9,13,14, 16,17,21, 22,24,25, 27,28,33, 34,36,37, 39	29	"	"	"	"	"

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TABLE XX. Output crosstalk (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S7	7B Output crosstalk, low level	5	8,9,13,14, 16,17,21, 22,24,25, 27,28,33, 34,36,37, 39	26	Open	0	800	mV	19,20
S8	8B "	27	5,8,9,13, 14,16,17, 21,22,24, 25,28,33, 34,36,37, 39	6	"	"	"	"	"
S9	9B "	33	5,8,9,13, 14,16,17, 21,22,24, 25,27,28, 34,36,37, 39	12	"	"	"	"	"
S10	10B "	13	5,8,9,14, 16,17,21, 22,24,25, 27,28,33, 34,36,37, 39	32	"	"	"	"	"
S11	11B "	9	5,8,13,14, 16,17,21, 22,24,25, 27,28,33, 34,36,37, 39	30	"	"	"	"	"
S12	12B "	34	5,8,9,13, 14,16,17, 21,22,24, 25,27,28, 33,36,37, 39	31	"	"	"	"	"

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TABLE XX. Output crosstalk (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S13	13B Output crosstalk, low level	14	5,8,9,13, 16,17,21, 22,24,25, 27,28,33, 34,36,37, 39	35	Open	0	800	mV	19,20
S14	14B "	16	5,8,9,13, 14,17,21, 22,24,25, 27,28,33, 34,36,37, 39	15	"	"	"	"	"
S15	15B "	36	5,8,9,13, 14,16,17, 21,22,24, 25,27,28, 33,34,37, 39	19	"	"	"	"	"
S16	16B "	39	5,8,9,13, 14,16,17, 21,22,24, 25,27,28, 33,34,36, 37	40	"	"	"	"	"
S17	17B "	37	5,8,9,13, 14,16,17, 21,22,24, 25,27,28, 33,34,36, 39	18	"	"	"	"	"
S18	18B "	17	5,8,9,13, 14,16,21, 22,24,25, 27,28,33, 34,36,37, 39	38	"	"	"	"	"

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TABLE XX. Output crosstalk (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S19	1B Output crosstalk, low level	21,22,24,25	5,8,9,13, 14,16,17, 27,28,33, 34,36,37, 39	3	Open	0	500	mV	19,20
S20	2B "	"	"	4	"	"	"	"	"
S21	3B "	"	"	2	"	"	"	"	"
S22	4B "	"	"	23	"	"	"	"	"
S23	5B "	5,8,27,28	9,13,14, 16,17,21, 22,24,25, 33,34,36, 37,39	7	"	"	"	"	"
S24	6B "	"	"	29	"	"	"	"	"
S25	7B "	"	"	26	"	"	"	"	"
S26	8B "	"	"	6	"	"	"	"	"
S27	9B "	9,13,33,34	5,8,14,16, 17,21,22, 24,25,27, 28,36,37, 39	12	"	"	"	"	"
S28	10B "	"	"	32	"	"	"	"	"
S29	11B "	"	"	30	"	"	"	"	"
S30	12B "	"	"	31	"	"	"	"	"
S31	13B "	14,16,17,37	5,8,9,13, 21,22,24, 25,27,28, 33,34,36, 39	35	"	"	"	"	"

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TABLE XX. Output crosstalk (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S32	14B Output crosstalk, low level	14,16,17,37	5,8,9,13, 21,22,24, 25,27,28, 33,34,36, 39	15	Open	0	500	mV	19,20
S33	15B "	17,36,37,39	5,8,9,13, 14,16,21, 22,24,25, 27,28,33, 34	19	"	"	"	"	"
S34	16B "	"	"	40	"	"	"	"	"
S35	17B "	"	"	18	"	"	"	"	"
S36	18B "	"	"	38	"	"	"	"	"

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TABLE XXI. Output crosstalk (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	1B Output crosstalk, high level	Open	5,8,9,13, 14,16,17, 21,22,24, 27,28,33, 34,36,37, 39	3	25	0	400	mV	19,20
T2	2B "	"	5,8,9,13, 14,16,17, 21,22,25, 27,28,33, 34,36,37, 39	4	24	"	"	"	"
T3	3B "	"	5,8,9,13, 14,16,17, 21,24,25, 27,28,33, 34,36,37, 39	2	22	"	"	"	"
T4	4B "	"	5,8,9,13, 14,16,17, 22,24,25, 27,28,33, 34,36,37, 39	23	21	"	"	"	"
T5	5B "	"	5,8,9,13, 14,16,17, 21,22,24, 25,27,33, 34,36,37, 39	7	28	"	"	"	"
T6	6B "	"	5,9,13, 14,16,17, 21,22,24, 25,27,28, 33,34,36, 37,39	29	8	"	"	"	"

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TABLE XXI. Output crosstalk (high-level) test - Continued.

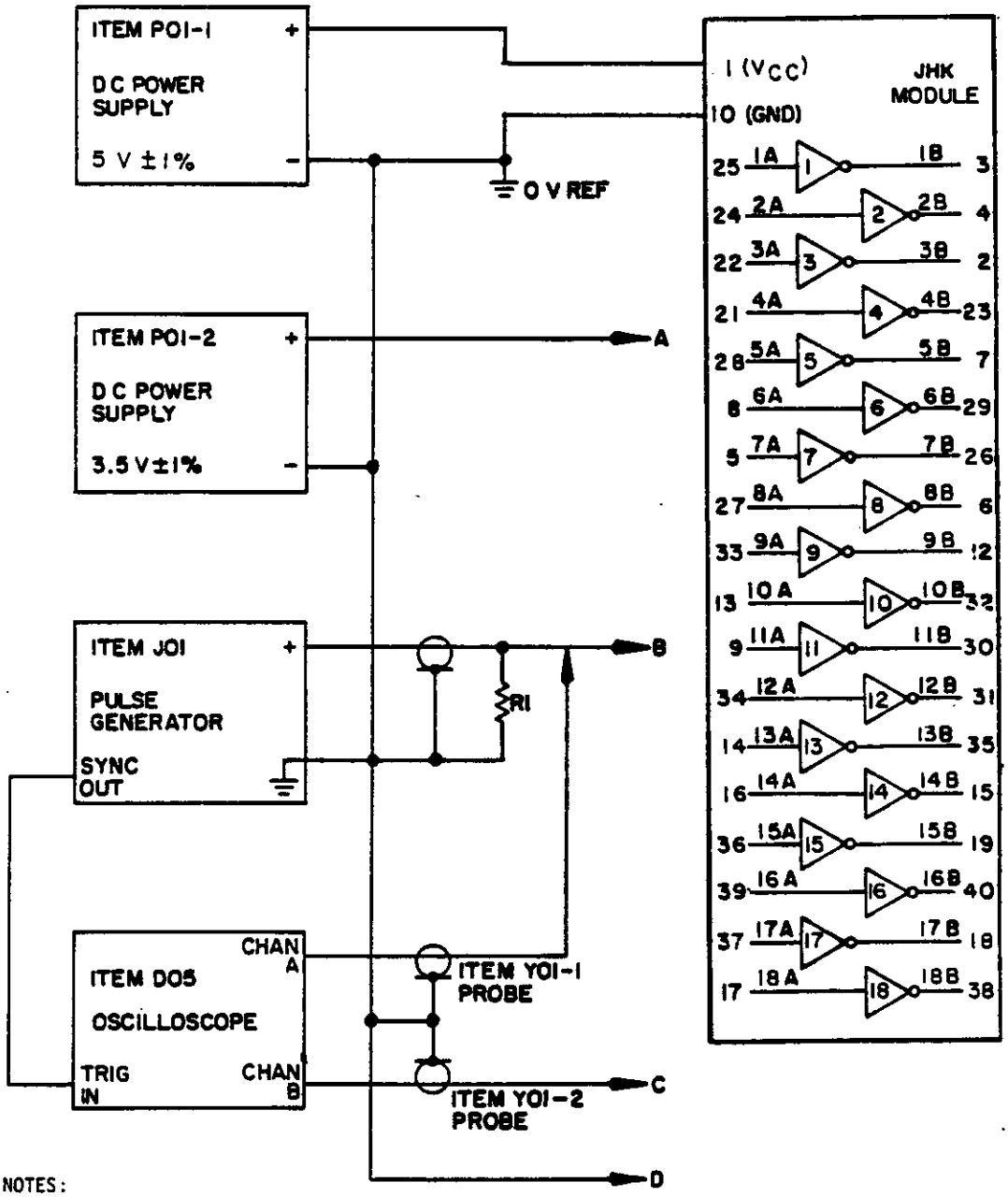
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T7	7B Output crosstalk, high level	Open	8,9,13,14, 16,17,21, 22,24,25, 27,28,33, 34,36,37, 39	26	5	0	400	mV	19,20
T8	8B "	"	5,8,9,13, 14,16,17, 21,22,24, 25,28,33, 34,36,37, 39	6	27	"	"	"	"
T9	9B "	"	5,8,9,13, 14,16,17, 21,22,24, 25,27,28, 34,36,37, 39	12	33	"	"	"	"
T10	10B "	"	5,8,9,14, 16,17,21, 22,24,25, 27,28,33, 34,36,37, 39	32	13	"	"	"	"
T11	11B "	"	5,8,13,14, 16,17,21, 22,24,25, 27,28,33, 34,36,37, 39	30	9	"	"	"	"
T12	12B "	"	5,8,9,13, 14,16,17, 21,22,24, 25,27,28, 33,36,37, 39	31	34	"	"	"	"

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TABLE XXI. Output crosstalk (high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T13	13B Output crosstalk, high level	Open	5,8,9,13, 16,17,21, 22,24,25, 27,28,33, 34,36,37, 39	35	14	0	400	mV	19,20
T14	14B "	"	5,8,9,13, 14,17,21, 22,24,25, 27,28,33, 34,36,37, 39	15	16	"	"	"	"
T15	15B "	"	5,8,9,13, 14,16,17, 21,22,24, 25,27,28, 33,34,37, 39	19	36	"	"	"	"
T16	16B "	"	5,8,9,13, 14,16,17, 21,22,24, 25,27,28, 33,34,36, 37	40	39	"	"	"	"
T17	17B "	"	5,8,9,13, 14,16,17, 21,22,24, 25,27,28, 33,34,36, 39	18	37	"	"	"	"
T18	18B "	"	5,8,9,13, 14,16,21, 22,24,25, 27,28,33, 34,36,37, 39	38	17	"	"	"	"

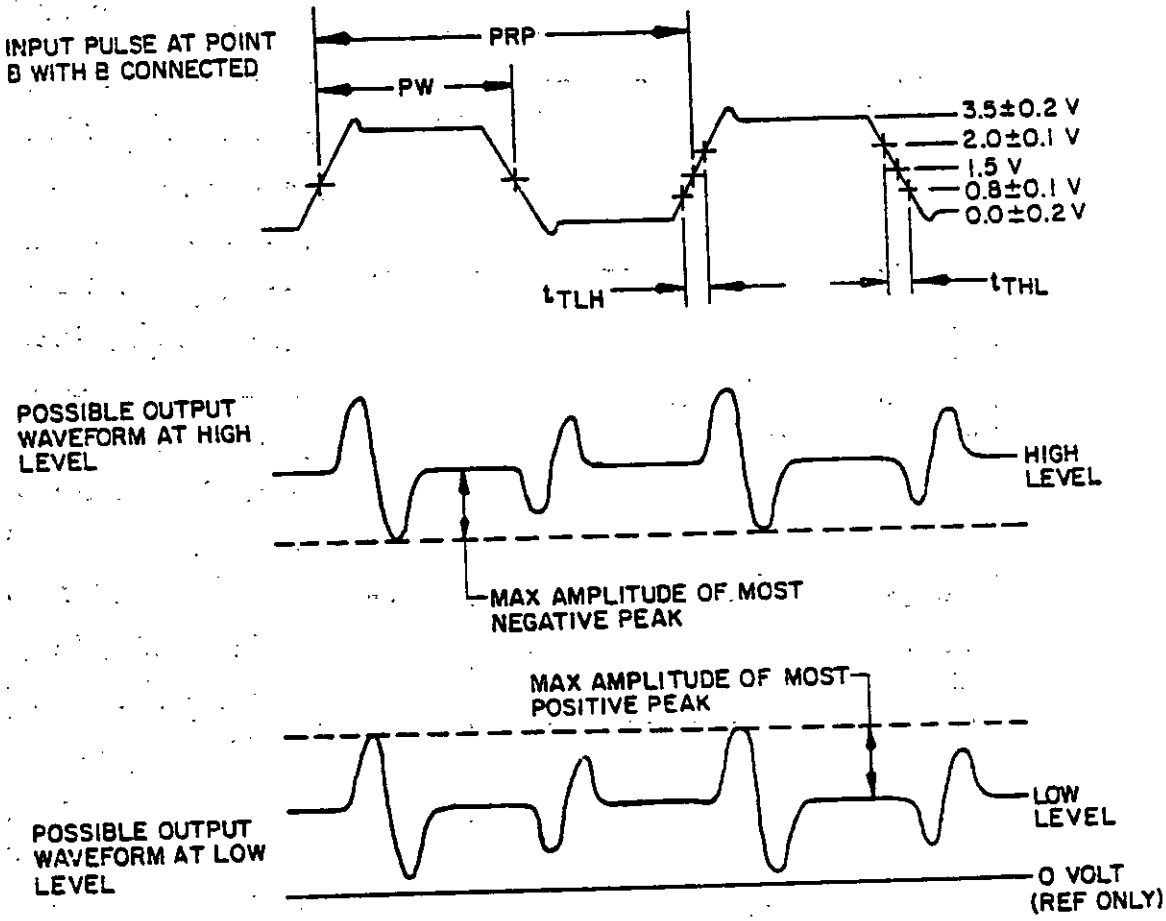
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NOTES:
1. RI = 51.1Ω ±1%, 1/4 W.
2. Load all outputs with load as specified on figure 12. Capacitor C_L is 50 ±5 pF for test tables XX and XXI.

FIGURE 19. Output crosstalk (low-level and high-level) test circuit for tables XX and XXI.

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NOTES:
1. Input pulse parameters:
   $t_{THL} = t_{TLH}$ = 10 ±2 ns (verified at each input).
   PW = 250 ±25 ns.
   PRP = 500 ±50 ns.
2. The lead length of electrical conductors between resistor R1 and input pins; between probes YO1-1 and YO1-2 and input and output pins; between load circuits and output pins; and between ground leads of probes YO1-1 and YO1-2 and test fixture and module ground pin, shall be less than 1 inch (25.4 mm).

* FIGURE 20. Output crosstalk (low-level and high-level) test circuit waveforms for tables XX and XXI.

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TABLE XXII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
U	Power supply transient amplitude	0	400	mV	21,22

TABLE XXIII. Power supply filter capacitance test.

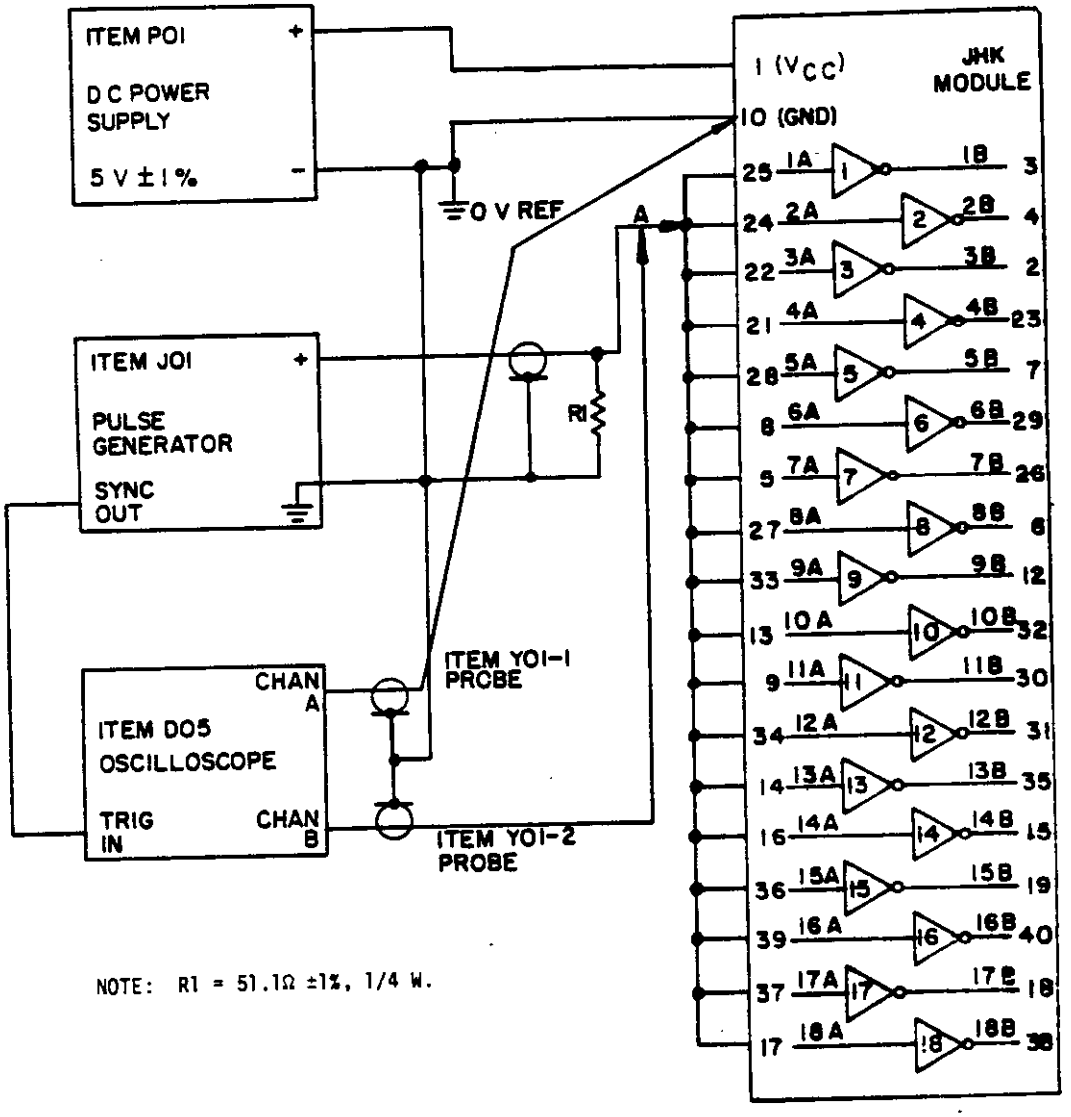
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V	Power supply filter capacitance	Open	1	1.6	3.0	µF	23

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TABLE XXIV. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W1	1A Input capacitance	25	Open	2	40	pF	23
W2	2A "	24	"	"	"	"	"
W3	3A "	22	"	"	"	"	"
W4	4A "	21	"	"	"	"	"
W5	5A "	28	"	"	"	"	"
W6	6A "	8	"	"	"	"	"
W7	7A "	5	"	"	"	"	"
W8	8A "	27	"	"	"	"	"
W9	9A "	33	"	"	"	"	"
W10	10A "	13	"	"	"	"	"
W11	11A "	9	"	"	"	"	"
W12	12A "	34	"	"	"	"	"
W13	13A "	14	"	"	"	"	"
W14	14A "	16	"	"	"	"	"
W15	15A "	36	"	"	"	"	"
W16	16A "	39	"	"	"	"	"
W17	17A "	37	"	"	"	"	"
W18	18A "	17	"	"	"	"	"

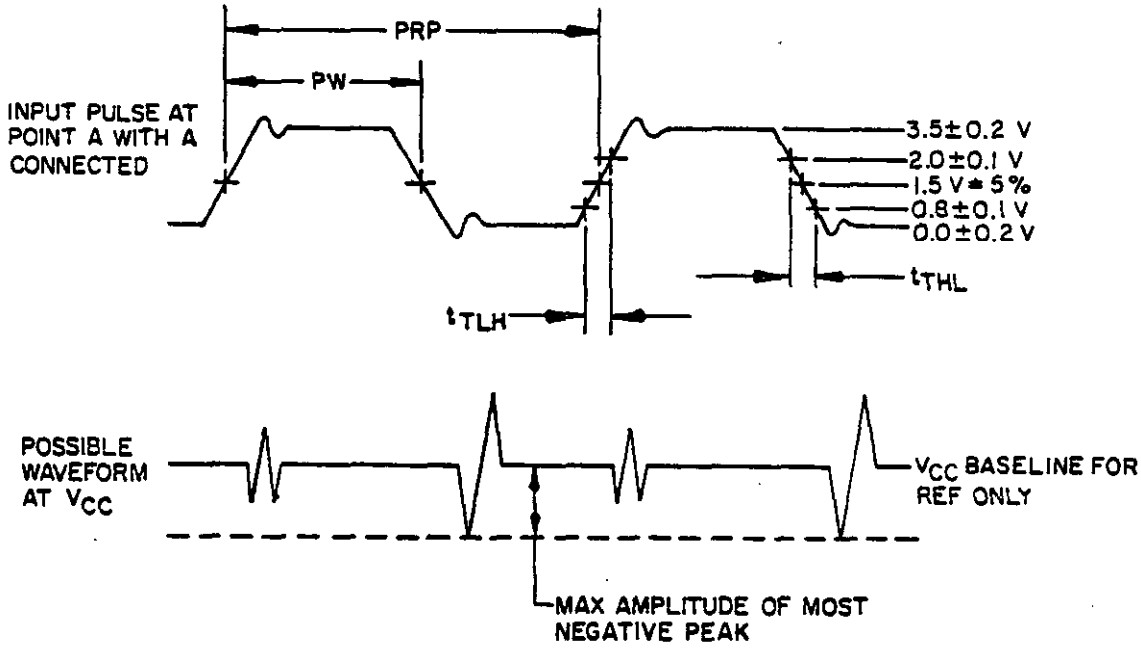
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NOTE: R1 = 51.1Ω ±1%, 1/4 W.

FIGURE 21. Power supply transient amplitude test circuit for table XXII.

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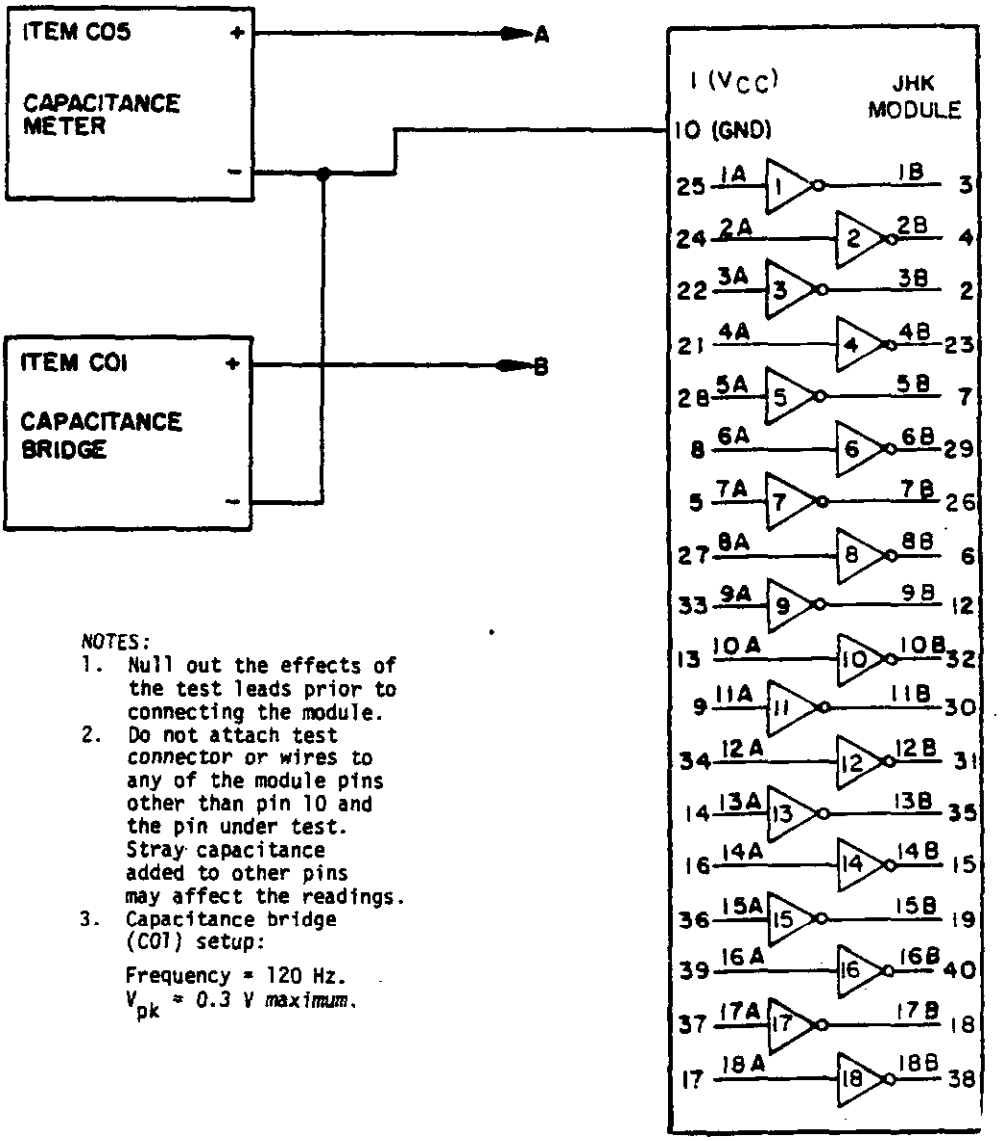


NOTE: Input pulse parameters:

$t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input)
PW = 500 ±50 ns.
PRP = 1,000 ±100 ns.

FIGURE 22. Power supply transient amplitude test circuit waveforms for table XXII.

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (C01) setup:

   Frequency = 120 Hz.
   V_{pk} = 0.3 V *maximum*.

FIGURE 23. Power supply filter capacitance and input capacitance test circuit for tables XXIII and XXIV.

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